

Speech and Audio Processing

08 - foundation model

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Class schedule (tentative)

01	Oct. 03	Introduction	Learn from basic to state-of-the-art about speech processing.
02	Oct. 10	Speech production / 音声生成	
03	Oct. 17	Speech synthesis / 音声合成	
04	Oct. 24	Speech synthesis 2 / 音声合成2	
05	Oct. 31	Speech recognition / 音声認識	
06	Nov. 7	Self-supervised learning / 自己教師あり学習	
07	Nov. 14	Sound recognition/synthesis / 環境音認識合成	
08	Nov. 28	Foundation model / 基盤モデル	
09	Dec. 05	LLM and dialogue system / LLMと対話システム	
10	Dec. 12	Speech system building / 音システム構築	Learn how to build your speech system.
11	Dec. 19	Speech system building / 音システム構築	
12	Dec. 26	Speech system building / 音システム構築	
13	Jan. 09	Speech system building / 音システム構築	
14	Jan. 23	Final presentation / 最終発表	

Previous class: special guest talk by Yuki Okamoto

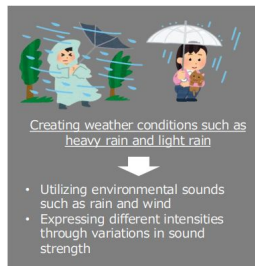
What is environmental sound?

■ Sounds that exist around us in daily life.

- e.g., car engine noise, alarm clock, rain, etc.
- Including speech and music

■ Environmental sounds are important in many situations.

- In media content such as animation and movies
 - ✓ To describe scenes
 - ✓ To express emotions of characters
- For understanding the surrounding environment
 - ✓ Humans detect hazards from sounds
 - Ex. 1: Hearing a car from behind
 - Be alert, a car is approaching
 - Ex. 2: A machine sounds different than usual
 - It may be malfunctioning.



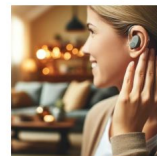
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What is environmental sound recognition?

■ A technology that enables computers to recognize a wide variety of sounds in our environment.

■ Applications

- AI assistants
 - ✓ Automatic life-logging
 - ✓ Autonomous driving
- Hearing assistance
 - ✓ Hearing aids
 - ✓ Automatic captioning
- Acoustic monitoring
 - ✓ Home care: baby caring, people fall over
 - ✓ Security: glass shattering, people screaming



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Environmental sound synthesis (ESS) / 環境音合成

■ Technology for artificially generating various environmental sounds.

- Recently, deep learning-based approaches have become mainstream.

■ Traditionally, research focused on synthesizing speech and music.

- Speech synthesis
- Singing voice synthesis
- Music synthesis



Applications :

- Generation of sound effects and background sounds for media content
- Realistic sound environments for VR and metaverse applications
 - ✓ Converting human voices into animal sounds, etc.
- Data augmentation for environmental sound recognition

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To learn more about environmental sound

■ DCASE (Detection and Classification and Acoustic Scenes and Events)

- International research community for environmental sound recognition and synthesis.
- Holds a **Workshop** and **Challenge** annually.

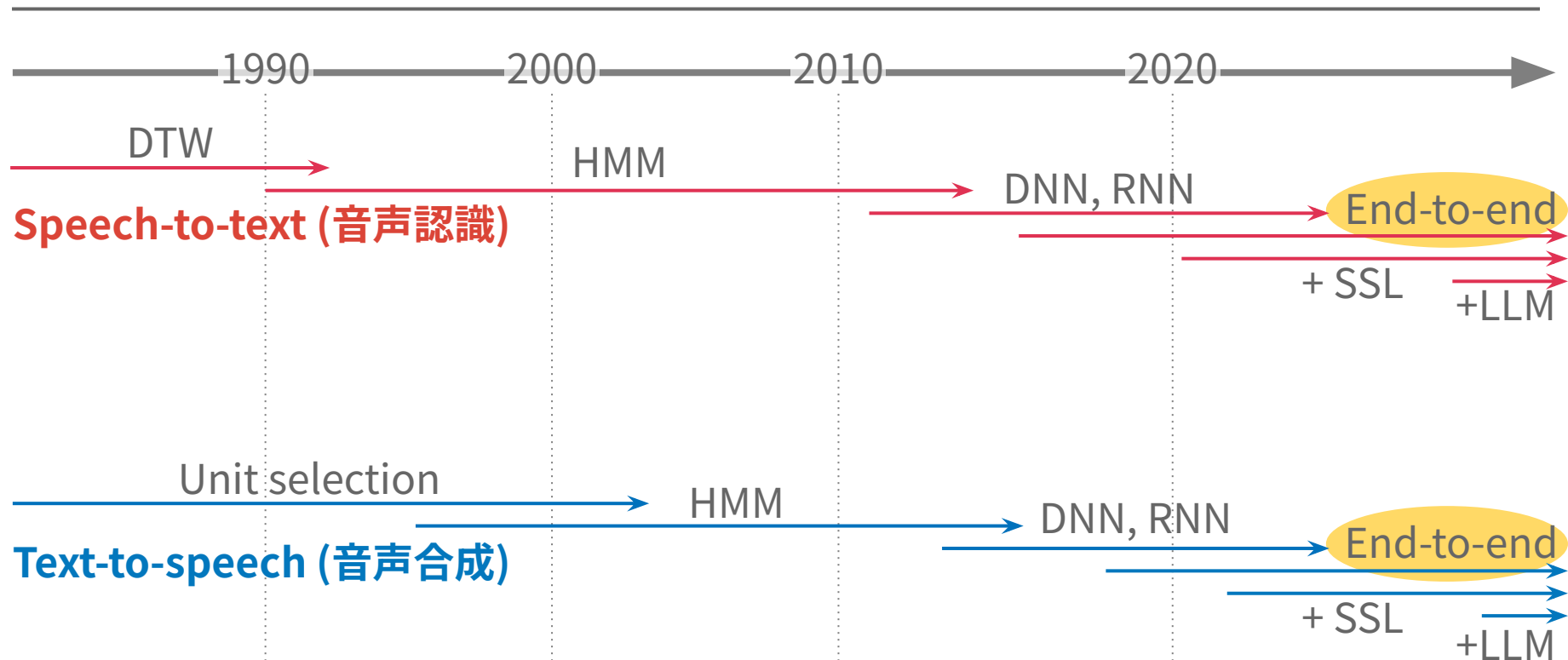
■ Publicly releases datasets, papers, and evaluation tools.

- Public release of papers presented at workshops
- Release of the datasets, results, and source code used in challenges
- Proposal of performance evaluation metrics and release of related tools



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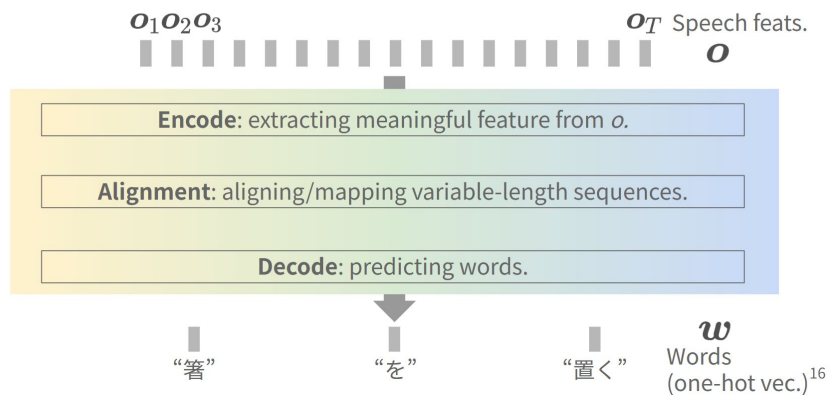
History of speech recognition and synthesis



End-to-end era: all modules are differentiable / 全てのモジュールが微分可能

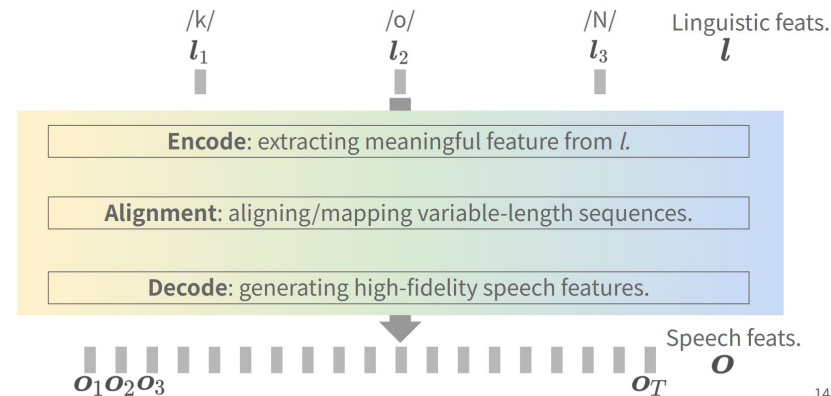
Speech-to-text (音声認識)

Into details: three inner modules

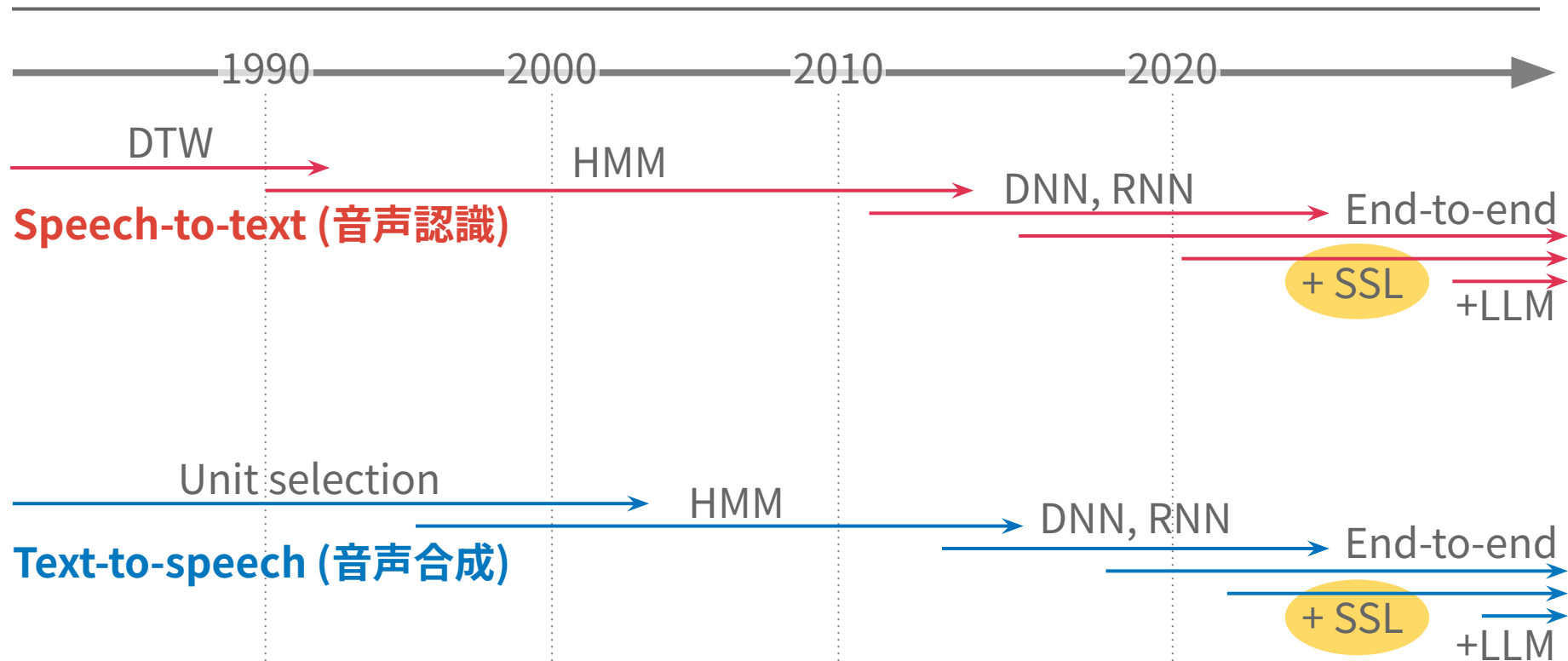


Text-to-speech (音声合成)

Into details: three inner modules

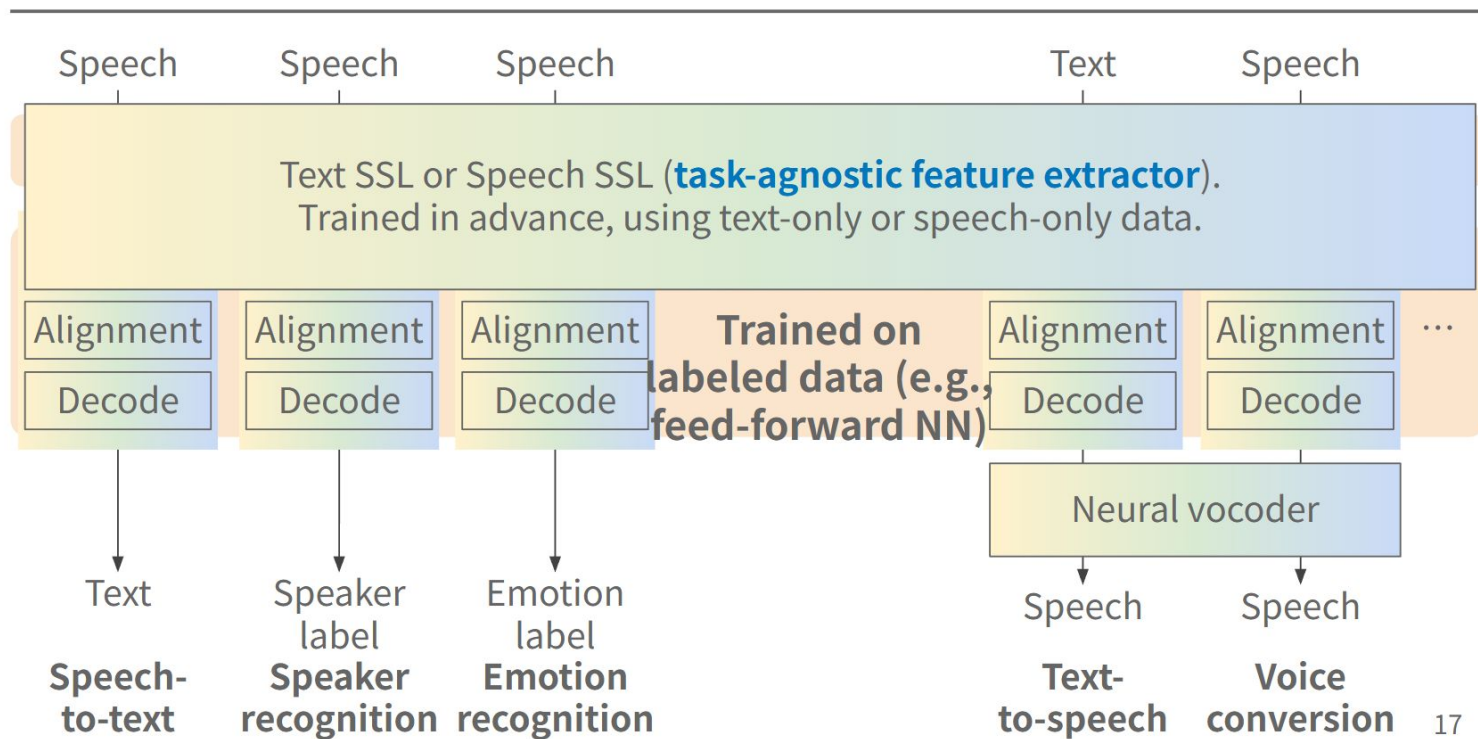


History of speech recognition and synthesis

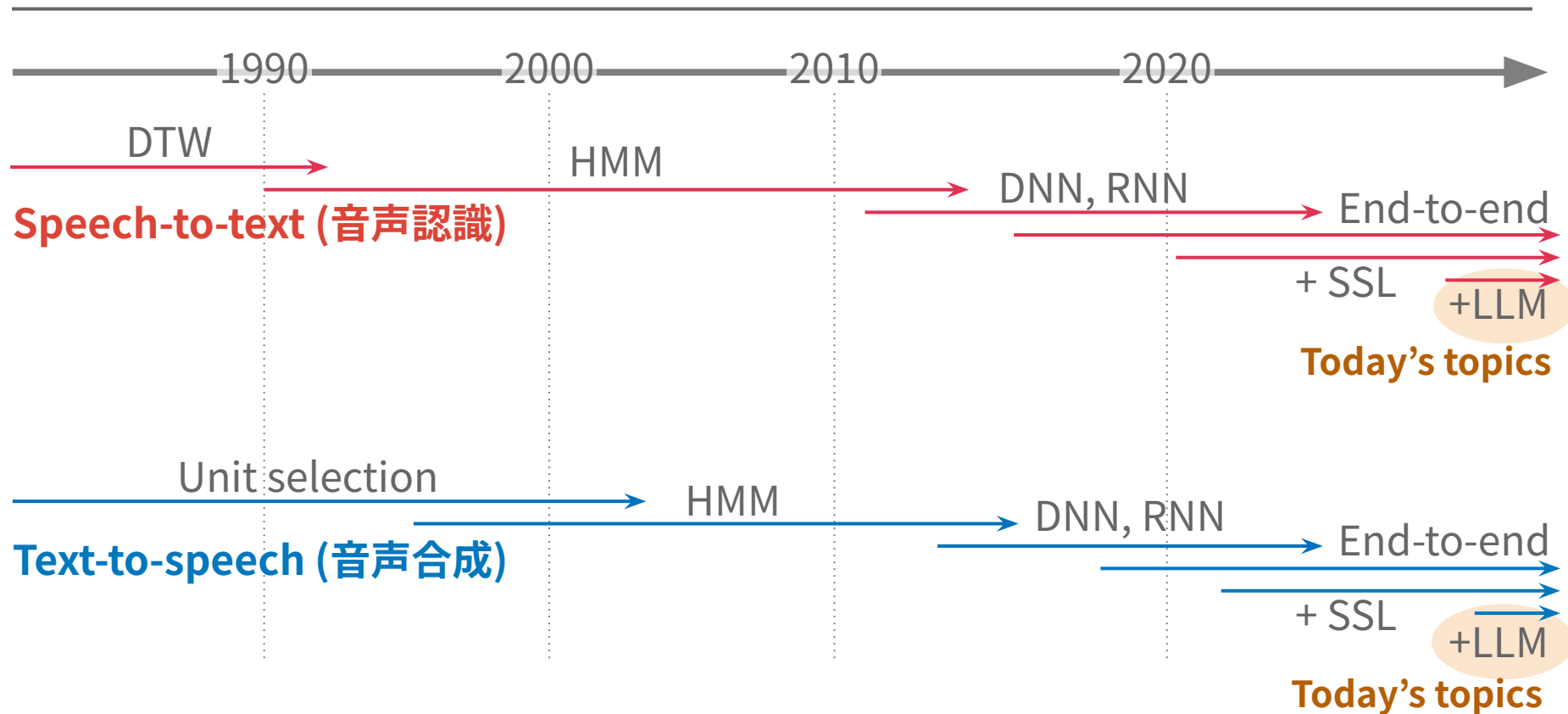


Self-supervised learning era: speech-only pretraining / 音声データだけで自己教師あり学習

SSL era

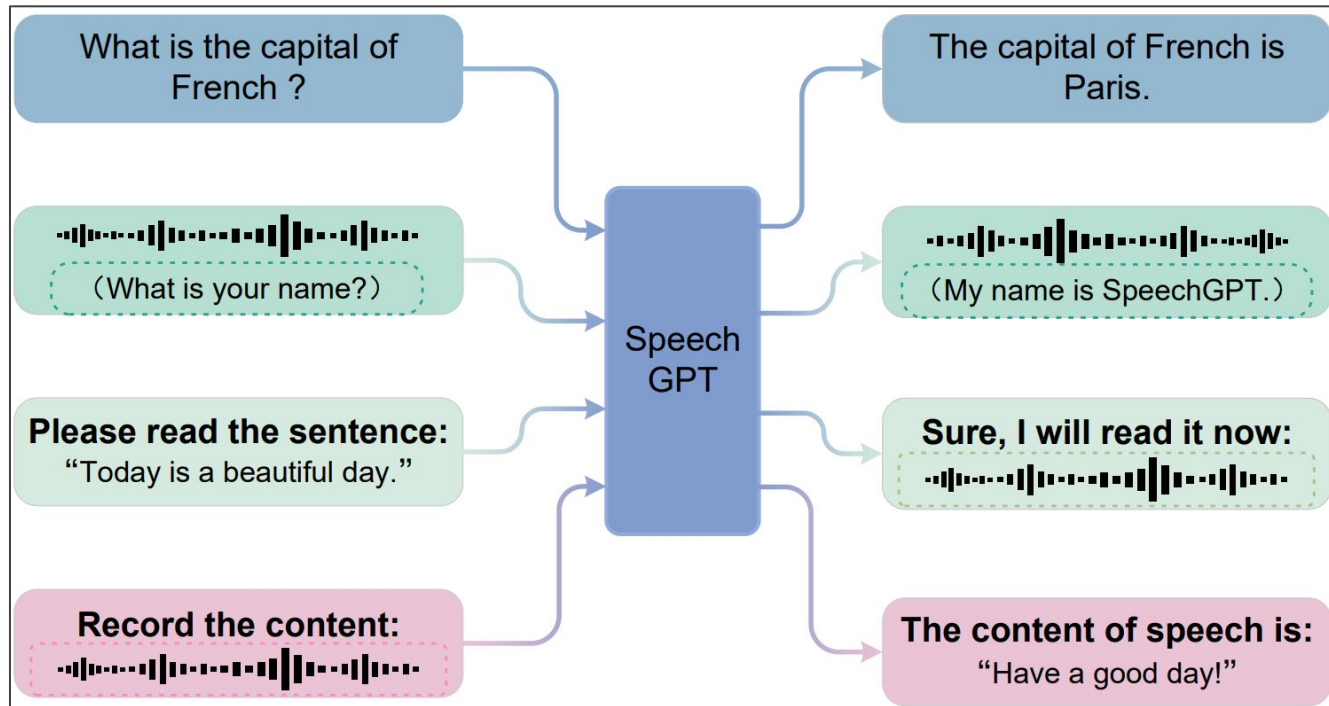


History of speech recognition and synthesis



Today's topic

Speech foundation model for all-in-one speech processing / あらゆる音声処理に対応する基盤モデル



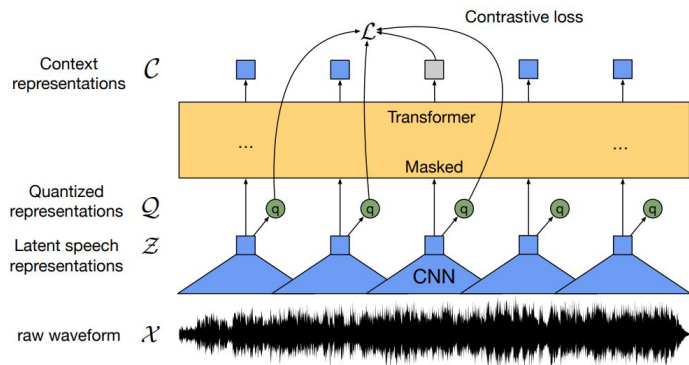
Two meanings of “foundation models” / “基盤モデル” の指す 2 つの意味

- “Large pre-trained models”
 - Self-supervised learning (SSL) model, e.g., HuBERT, wavLM.
 - Large supervised learning model, e.g., Whisper
- “All-in-one models” (**today’s main topic**)
 - Covers several tasks in one model.
 - Large language model, and spoken language model (Speech language model)
 - A.k.a.,
 - **spoken language model (SLM, 音声言語モデル),**
 - **speech LLM (音声モデル), or**
 - **speech foundation model (音声基盤モデル)**

“Large pre-trained models” / 大規模学習済みモデルとして

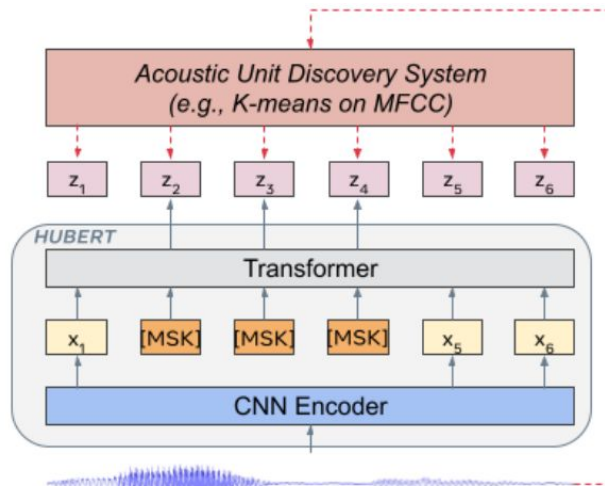
Self-supervised learning (SSL)

- “Self-supervised learning is a machine learning technique that uses unsupervised learning (...). Rather than relying on labeled datasets for supervisory signals, self-supervised models generate implicit labels from unstructured data.”
/ 自己教師あり学習とは, (中略) 教師なし学習を用いる機械学習手法であり, ラベル付きデータではなく, ラベル無しデータから生成した暗黙的なラベルを利用する.



<https://arxiv.org/pdf/2006.11477>

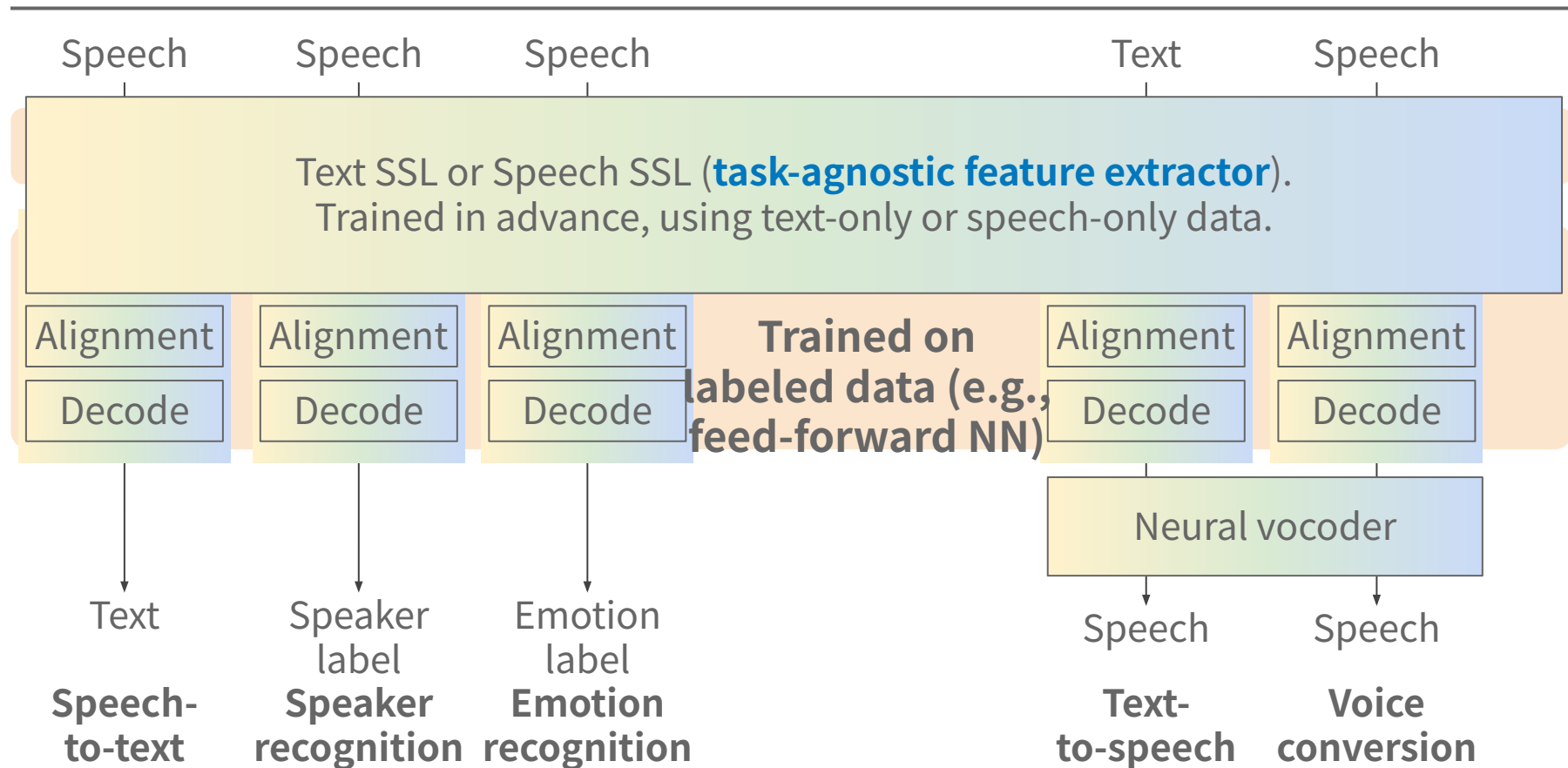
wav2vec 2.0



<https://arxiv.org/pdf/2106.07447>

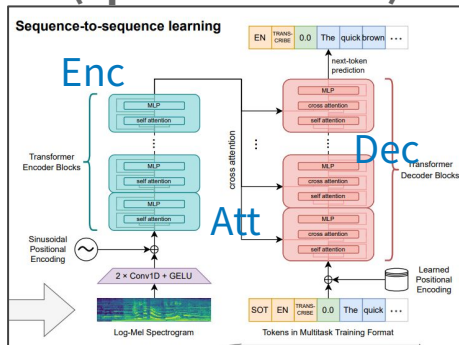
HuBERT

SSL models as universal speech feature extractor

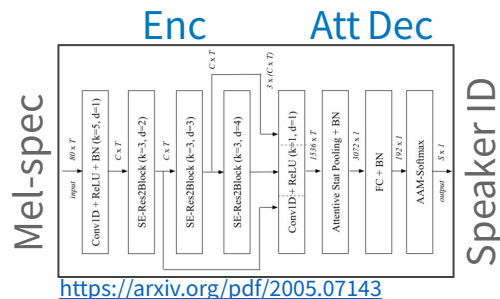


Large **supervised** models are also good at feature extraction.

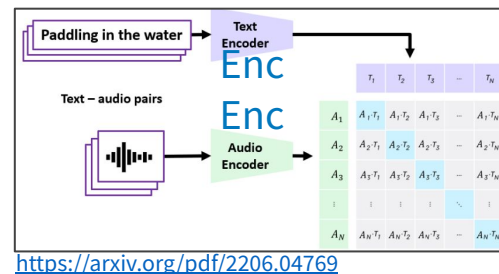
Whisper (speech-to-text)



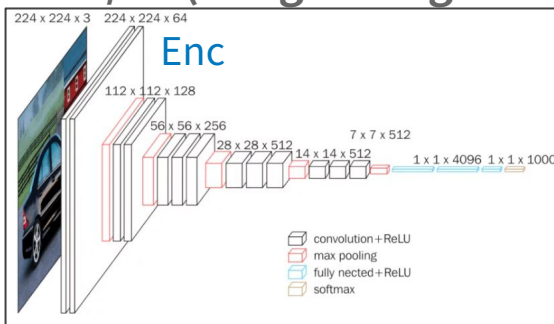
ECAPA-TDNN (speaker recognition)



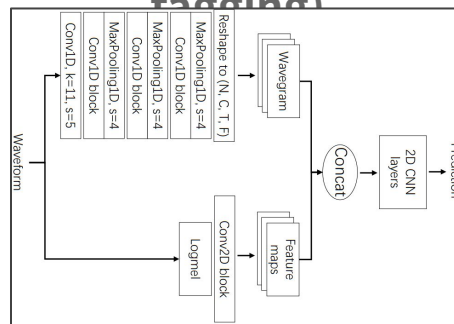
CLAP * audio CLIP (text-audio alignment)



VGG16/19 (image recognition)



PANNs (audio tagging)

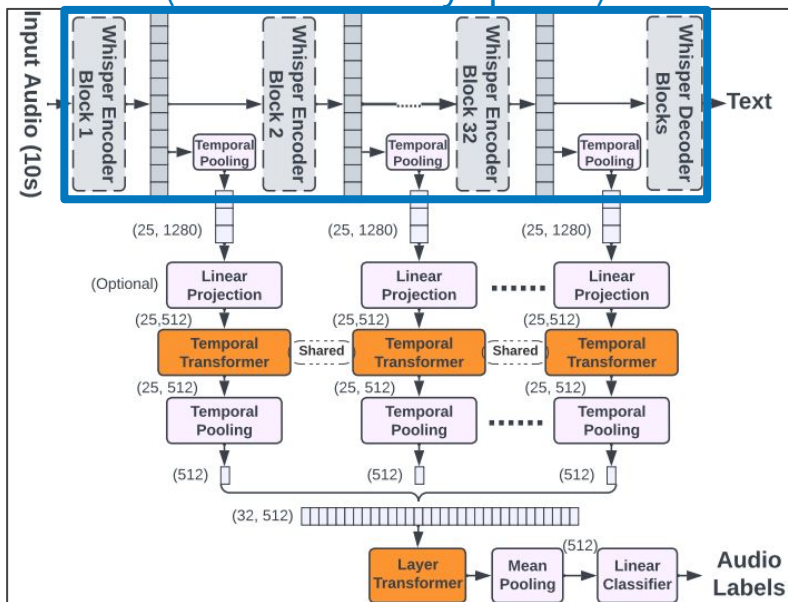


* spectrogram
as an image

Large **supervised** models are also good at feature extraction.

The **Whisper encoder** is a strong feature extractor for audio tagging.

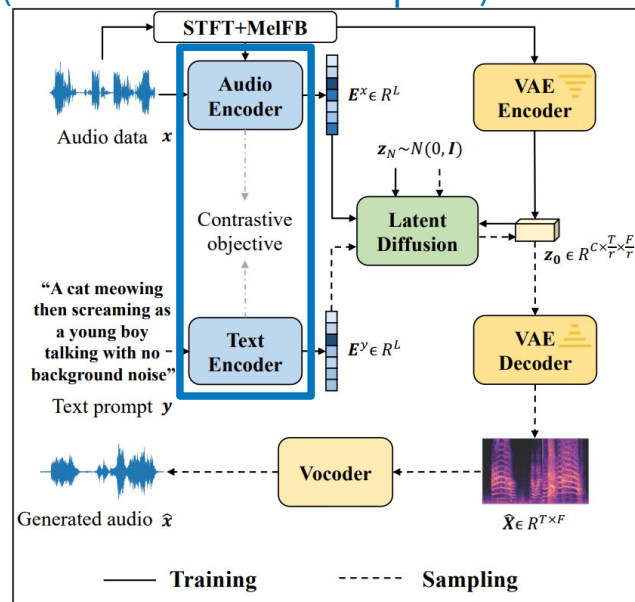
Pretrained Whisper encoder
(trained on noisy speech)



<https://arxiv.org/pdf/2307.03183>

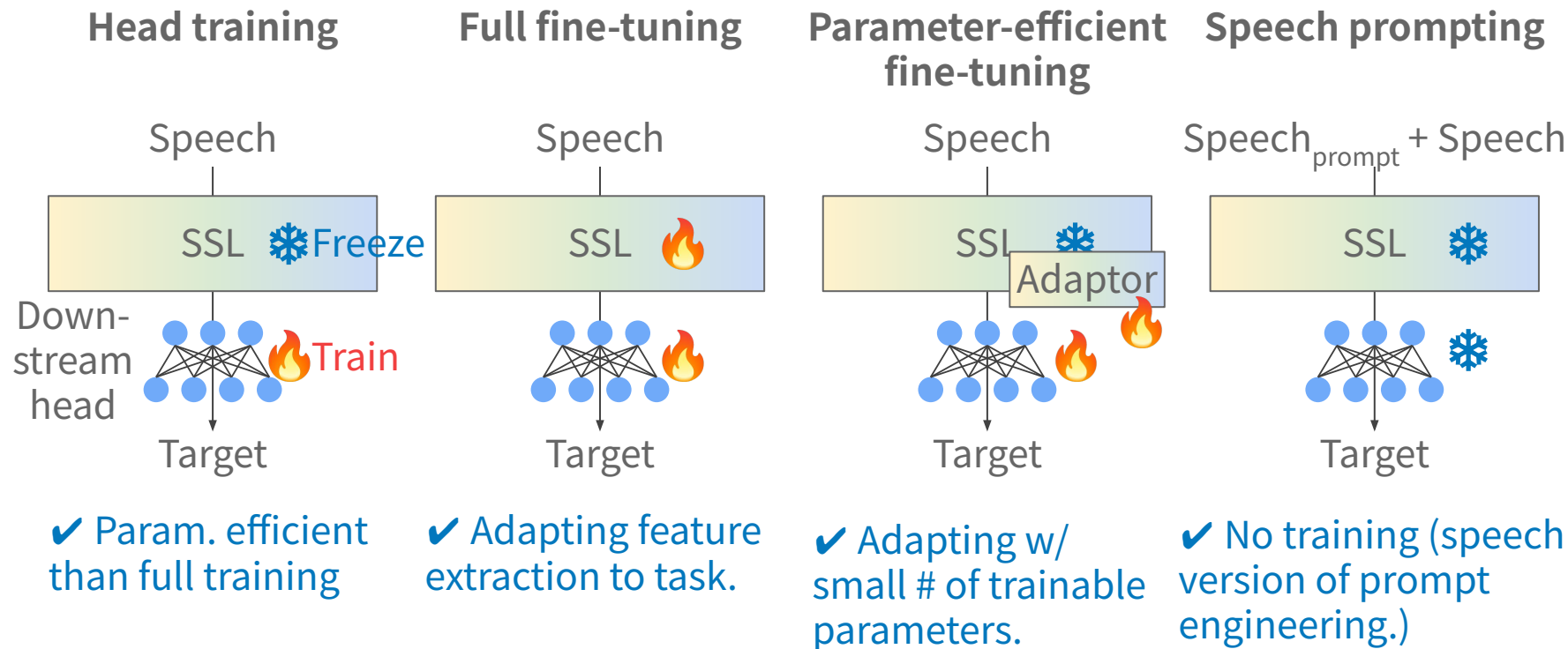
The **CLAP encoders** are strong feature extractors for text-to-audio generation.

Pretrained CLAP encoders
(trained on text-audio pairs)

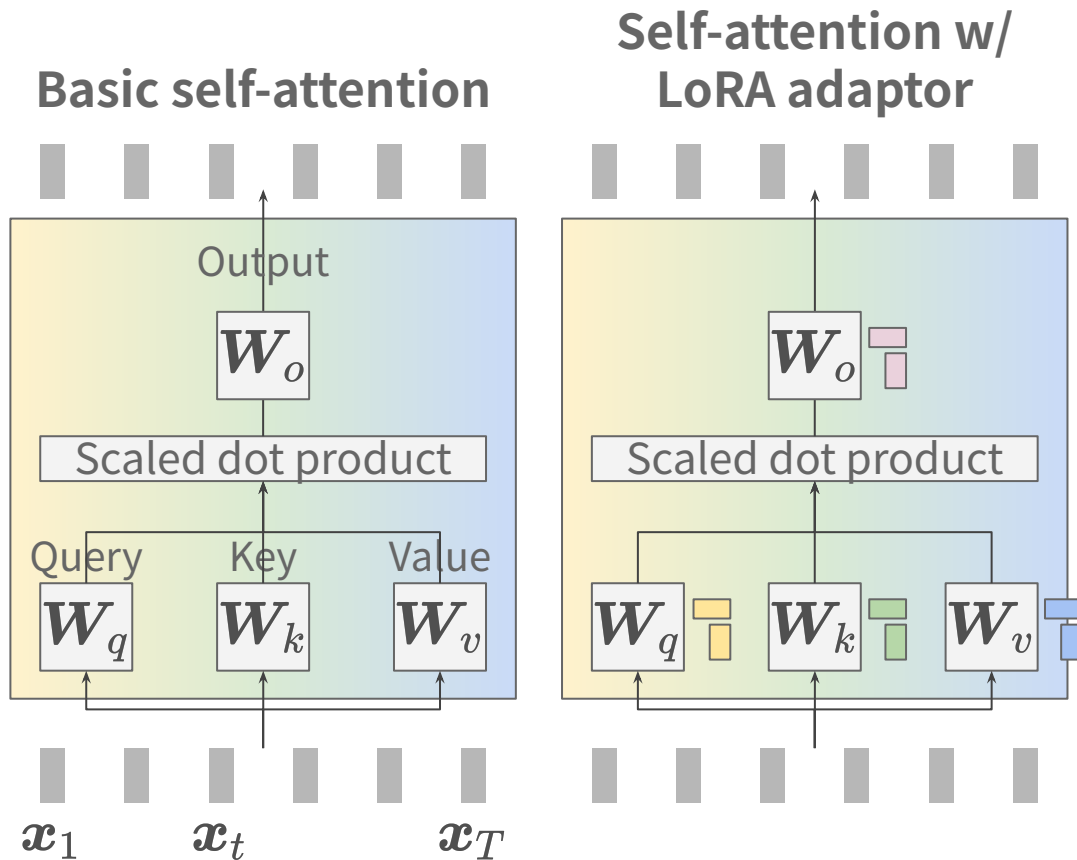


<https://arxiv.org/pdf/2301.12503>

Fine-tuning techniques (not only SSL but also other large supervised models)



LoRA: Low-rank adaptation



Basic finetuning

$$y = Wx$$

LoRA finetuning

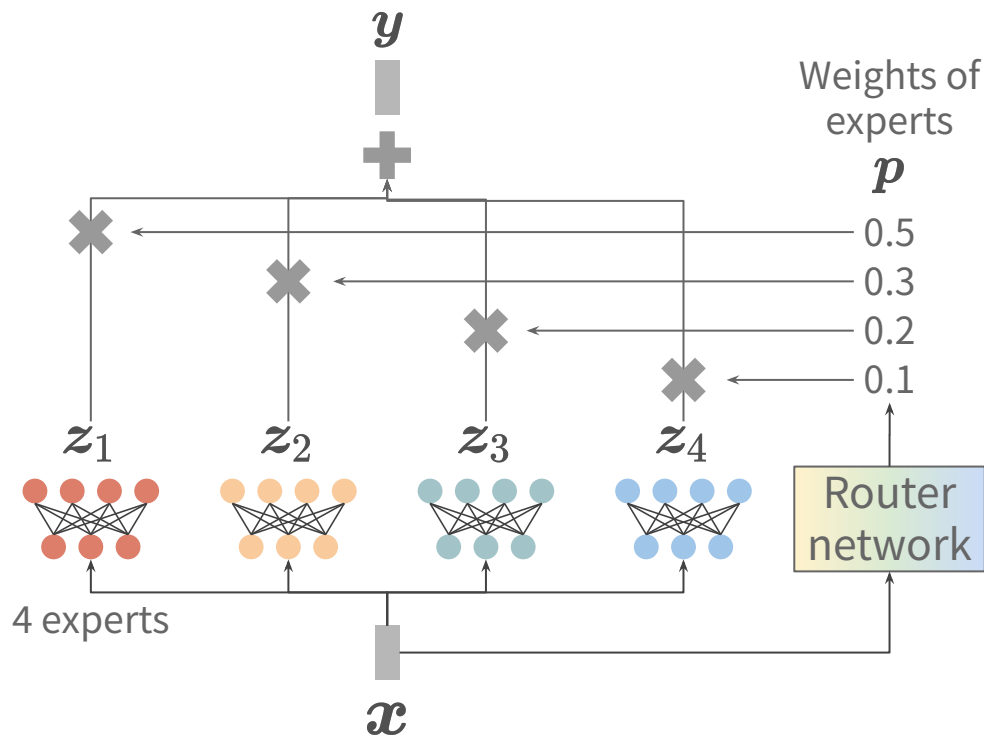
$$y = (W + \underline{AB})x$$

Updating parameters
of low-rank matrices

Mixture of experts (MoE) for neural networks

Instead of using one model, mixing outputs from multiple (small) expert models.

/ 1つのモデルの代わりに，複数の (小さな) 専門モデルの出力を加算する。



Basic one model

$$y = f(x)$$

Mixture of experts

$$y = \sum_{\text{all } i} p_i z_i = \sum_{\text{all } i} p_i f_i(x)$$

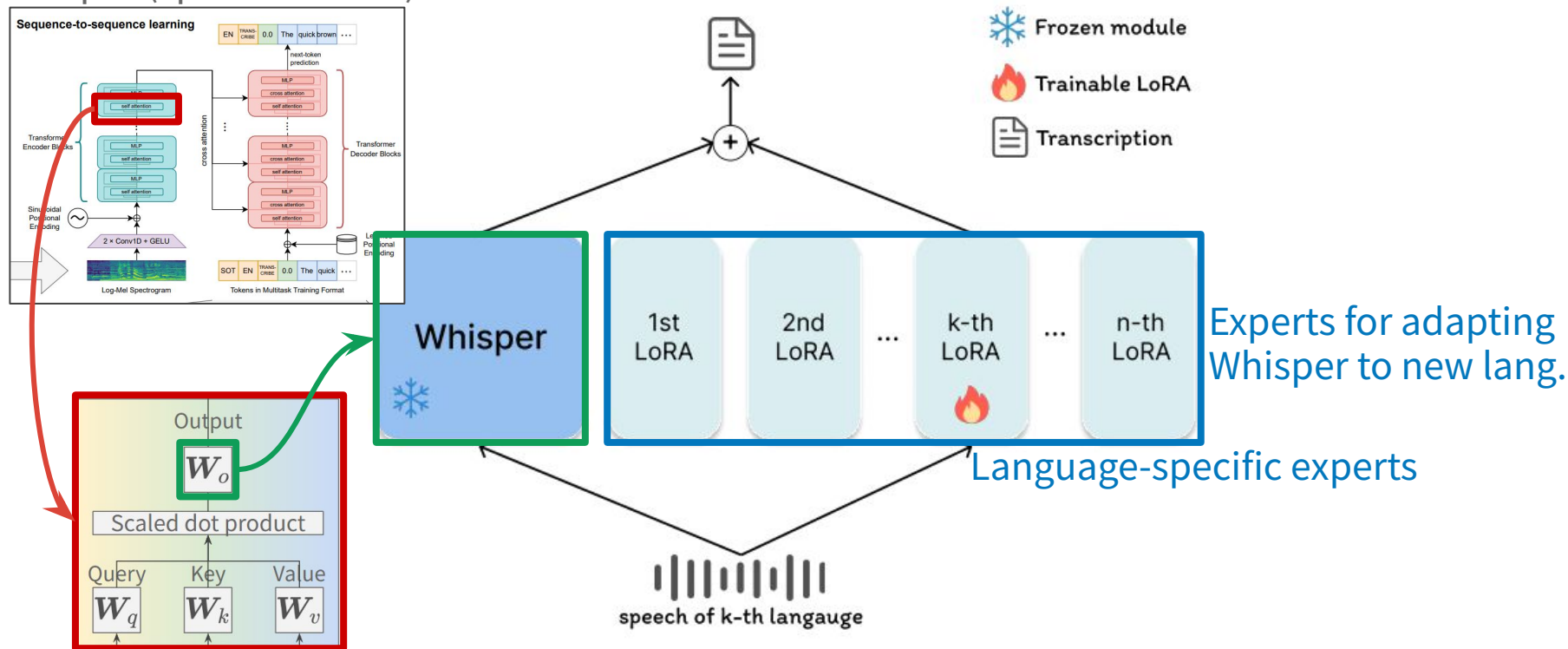
Output of i -th expert

When applying top- k algorithm, the highest k experts have non-zero weights.

MoE + LoRA for multi-lingual ASR

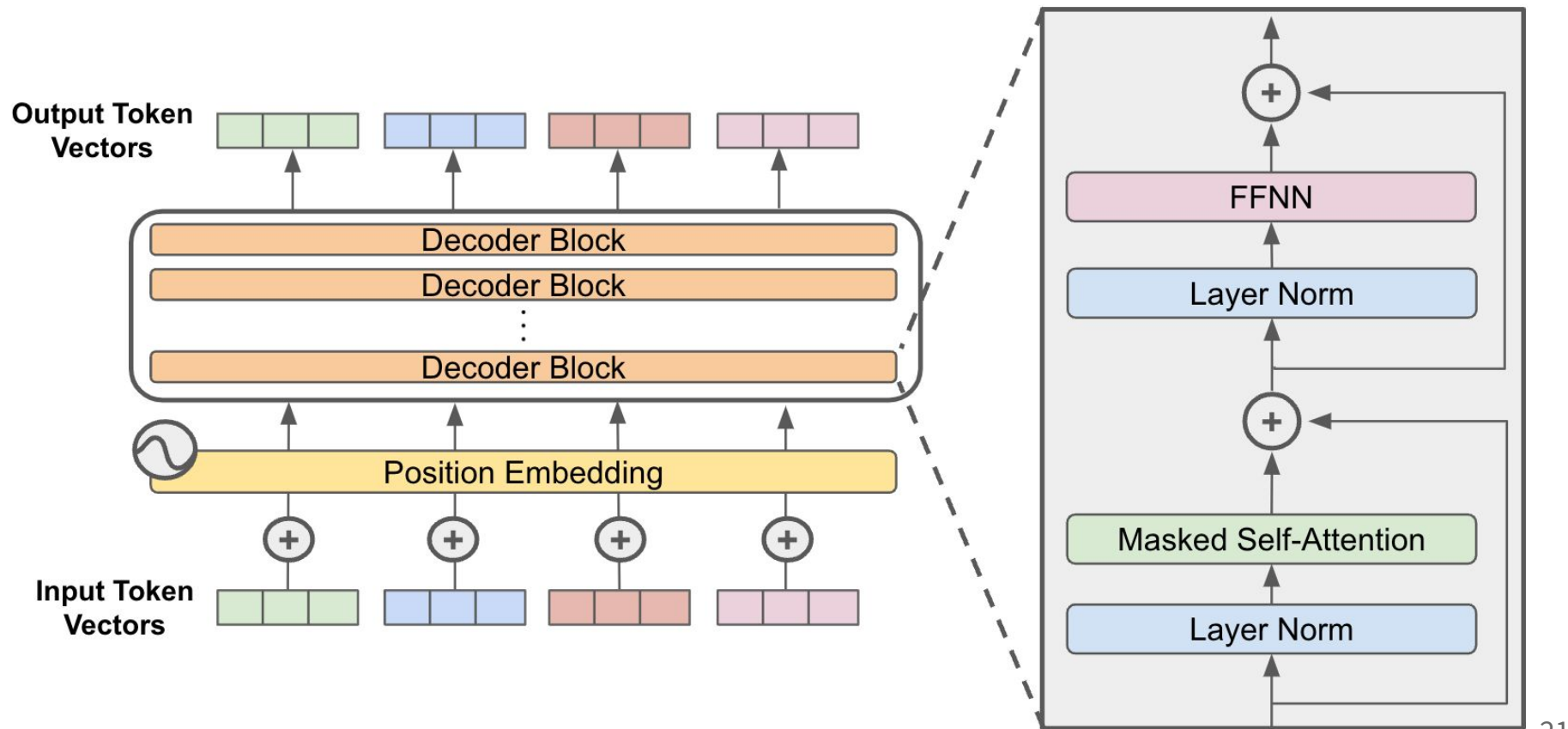
/ 新たな言語の音声認識のための MoE + LoRA

Whisper (speech-to-text)

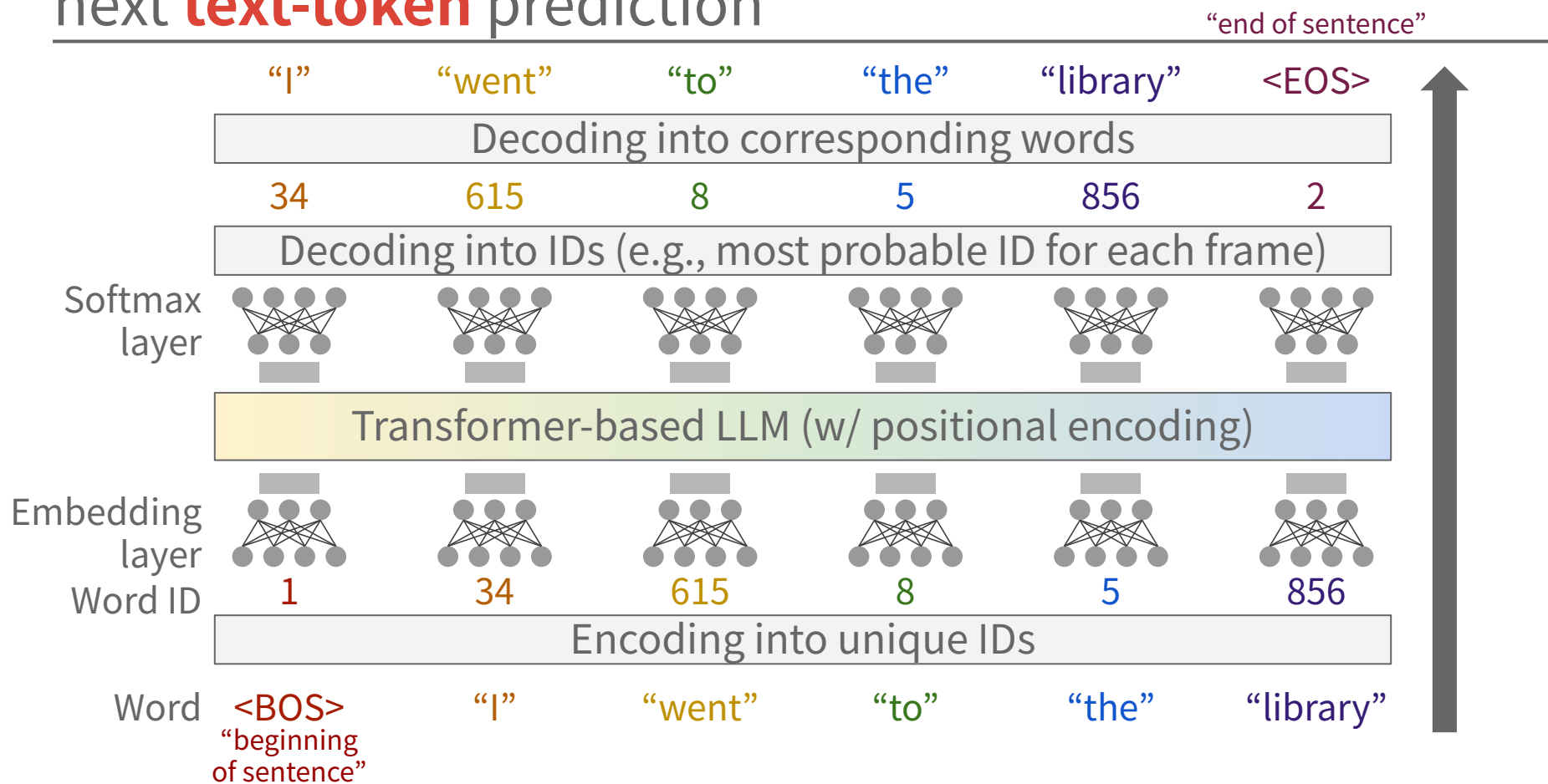


“All-in-one model” / あらゆるタスクに対応するモデル

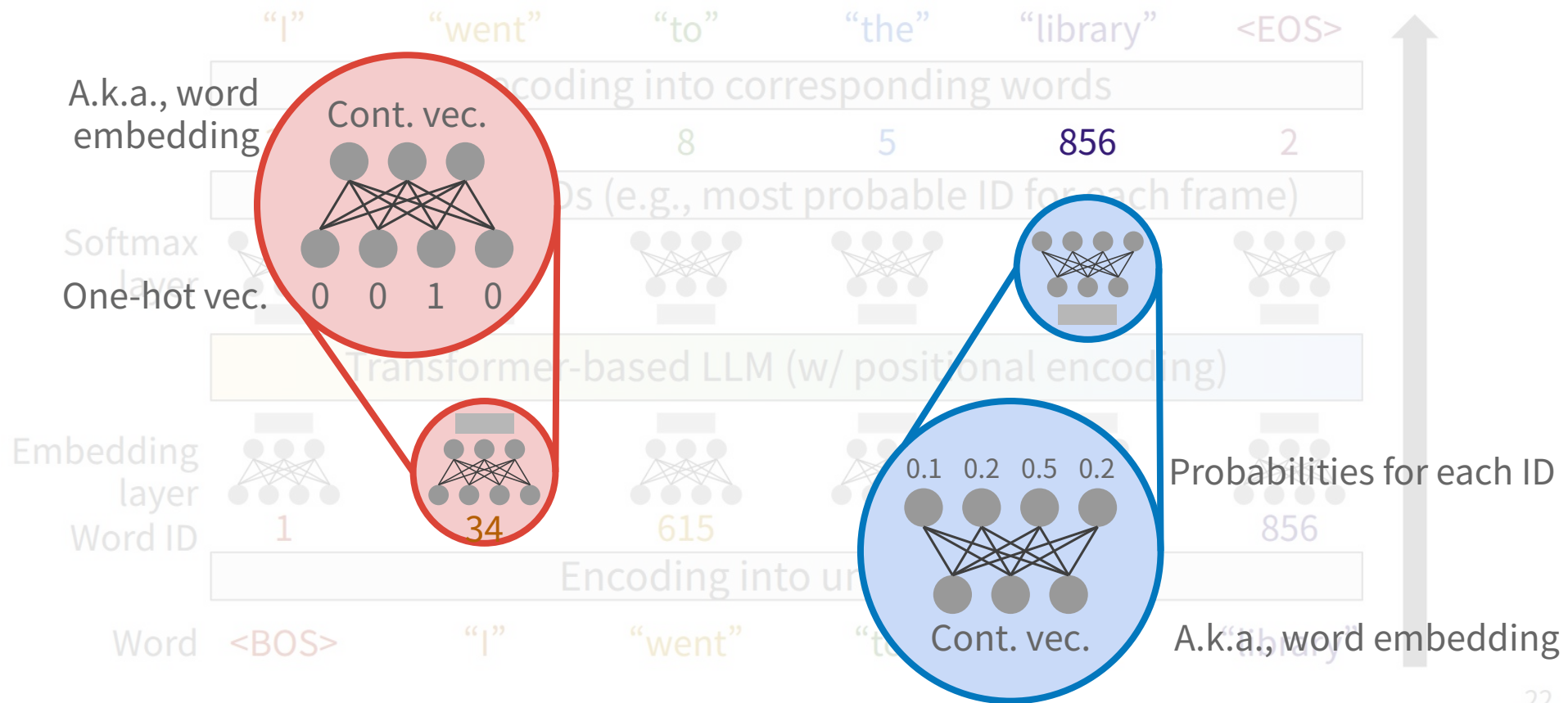
Backbone Transformer decoder



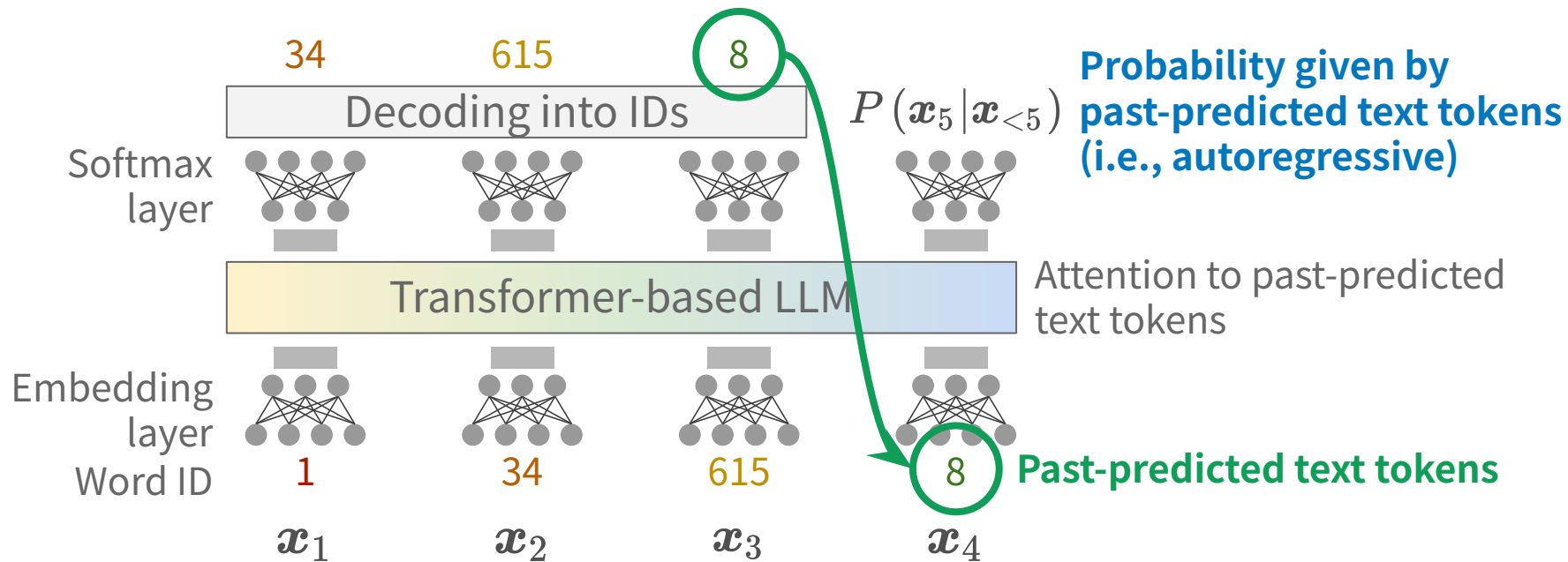
Transformer decoder-based LLM: next **text-token** prediction



Transformer decoder-based LLM: next **text-token** prediction



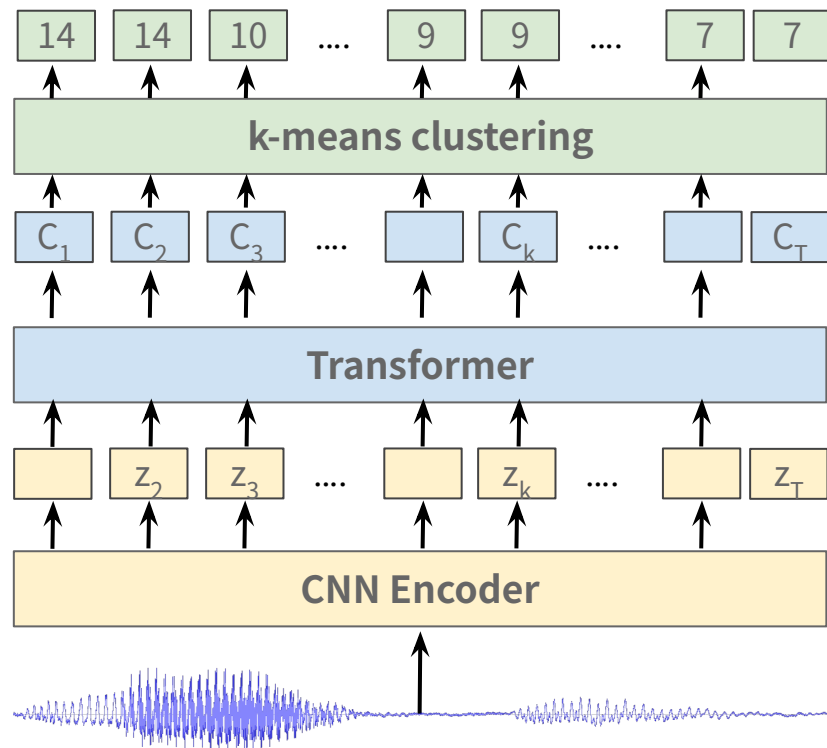
Inference of **text-tokens**



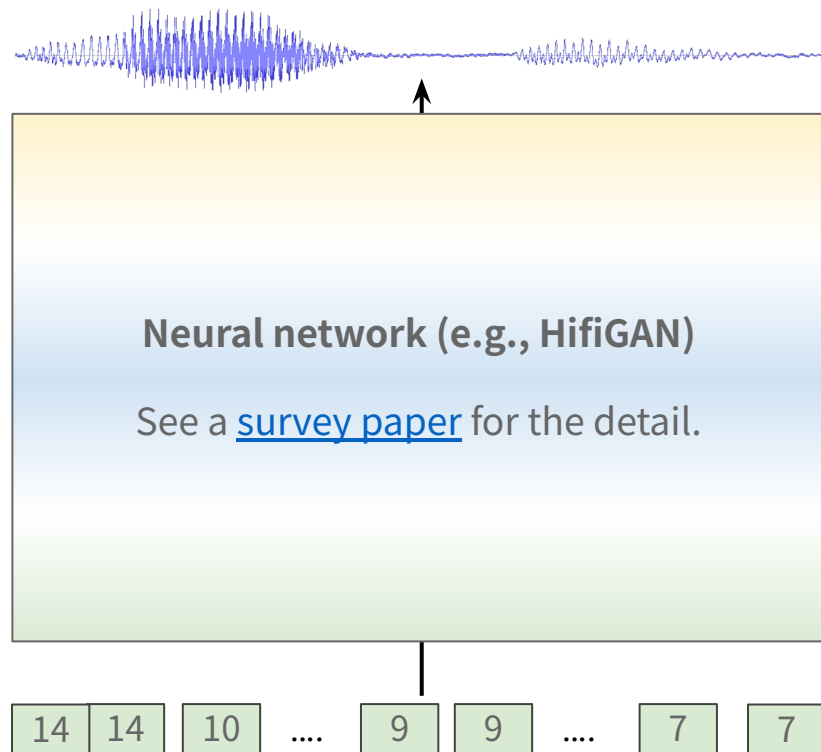
Case 1: Speech-only modeling

Speech encoder and decoder

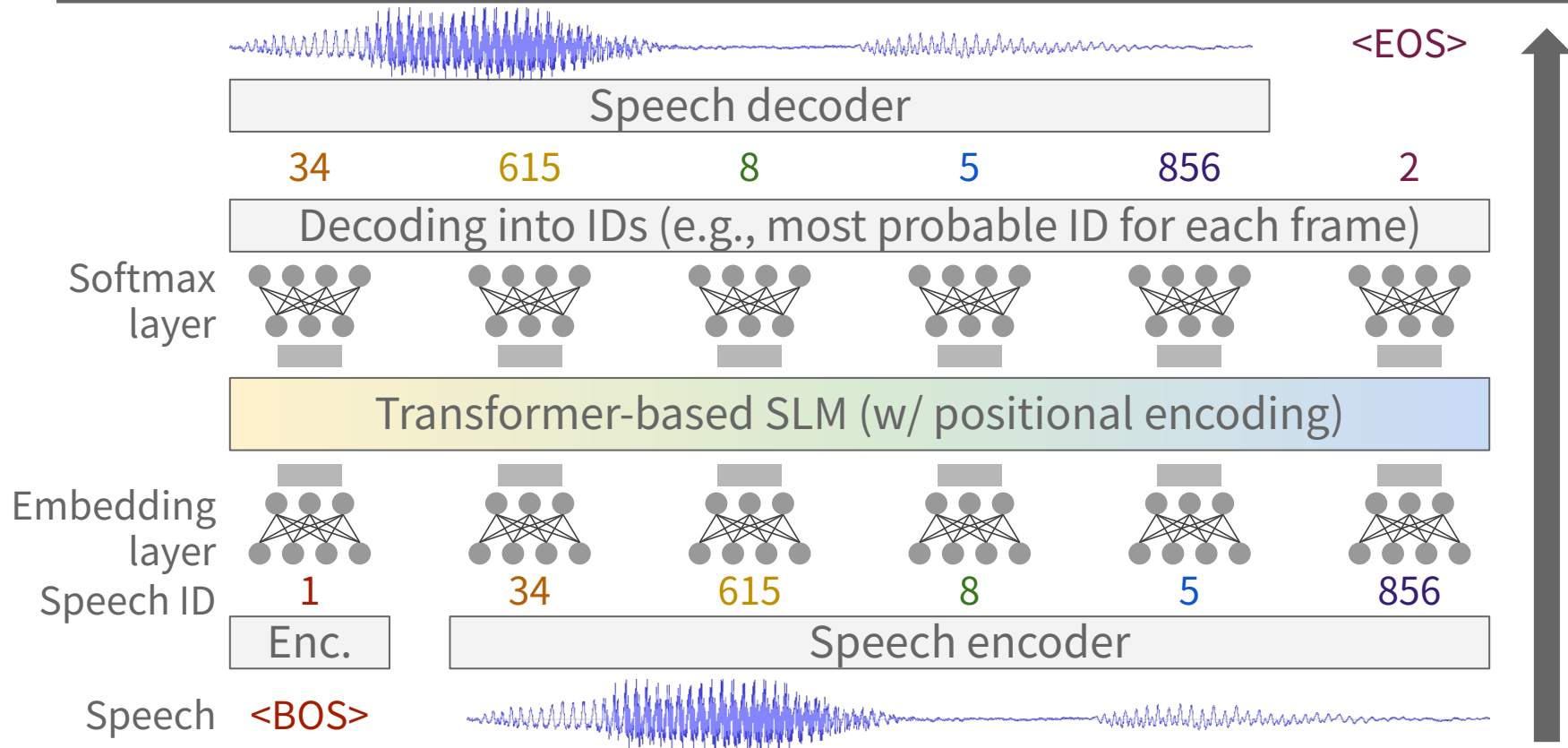
Encoder (pretrained SSL + k-means)



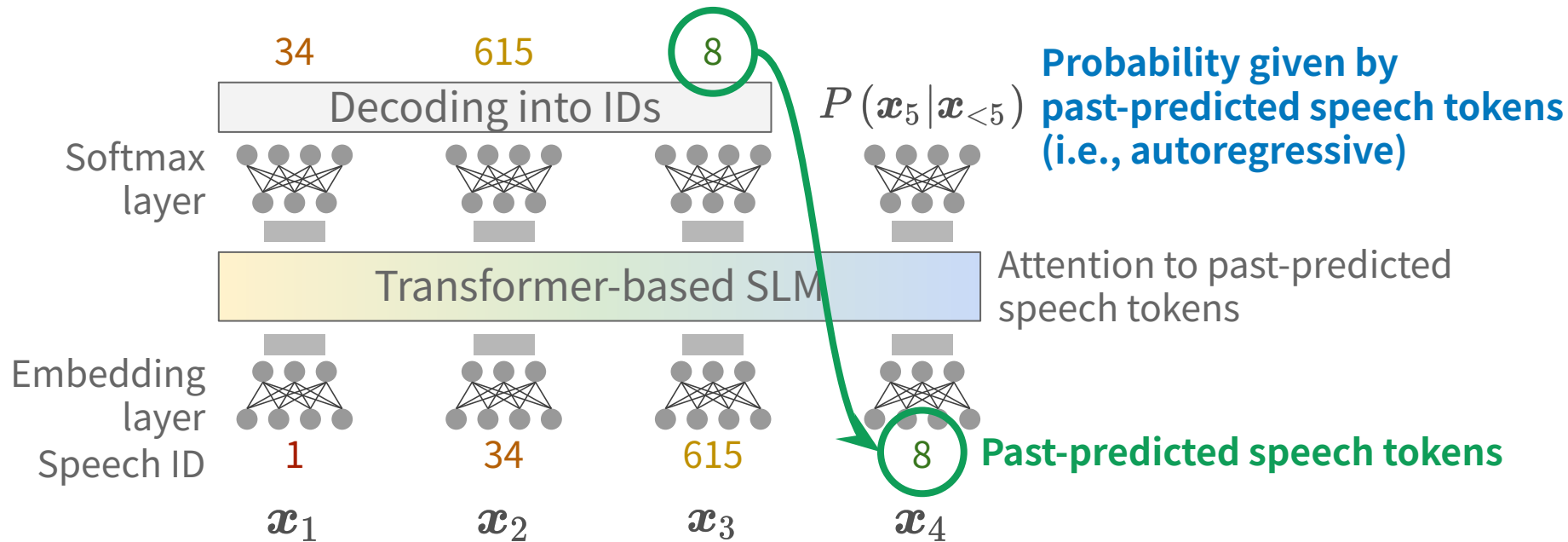
Decoder (neural vocoder)



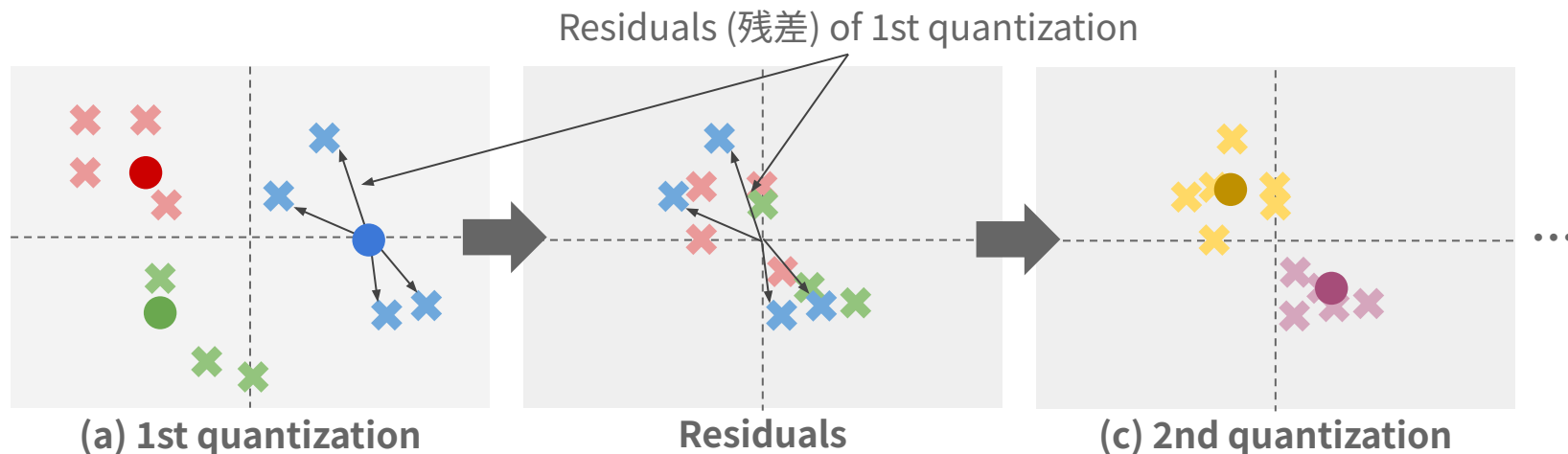
Speech-only modeling (next **speech-token** prediction)



Inference of **speech-tokens**



(Revisit) residual vector quantization (RVQ)



Algorithm 1: Residual Vector Quantization

Input: $y = \text{enc}(x)$ the output of the encoder, vector
quantizers Q_i for $i = 1..N_q$

Output: the quantized $\hat{y}$

$\hat{y} \leftarrow 0.0$

residual $\leftarrow y$

for $i = 1$ to N_q **do**

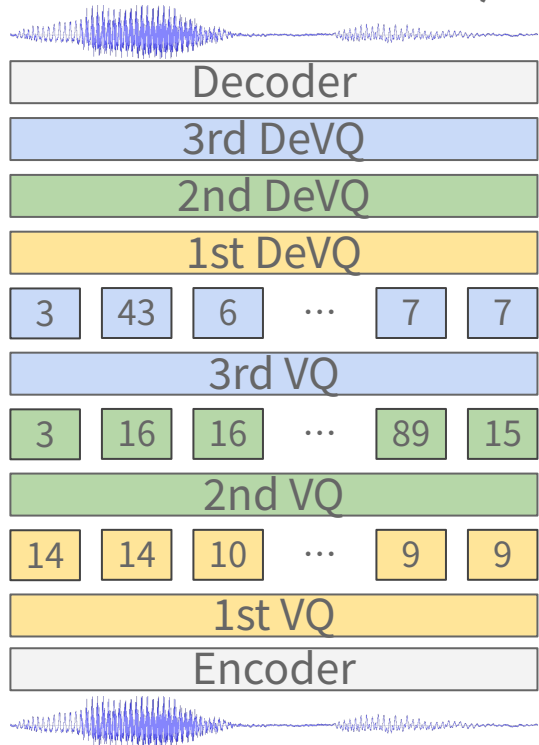
$\hat{y} += Q_i(\text{residual})$

 residual $-= Q_i(\text{residual})$

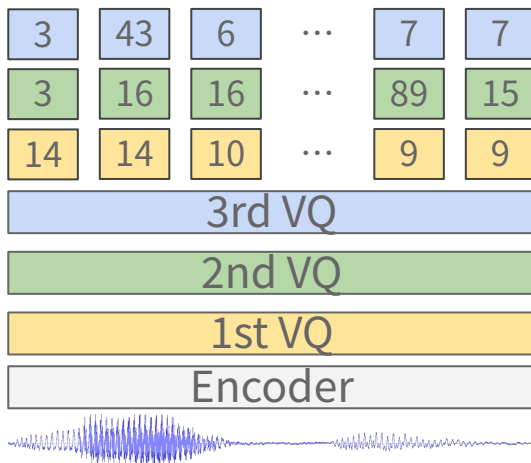
return $\hat{y}$

(Revisit) Soundstream: neural RVQ

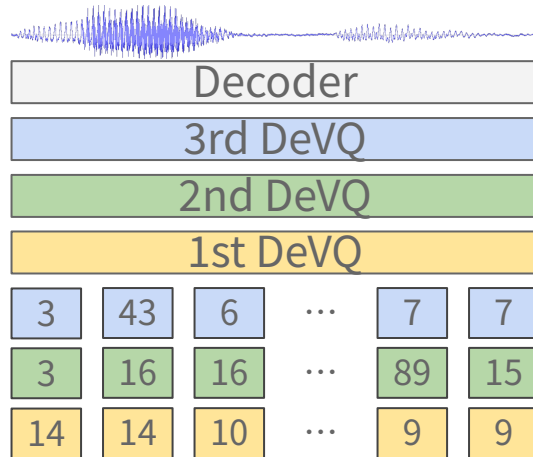
Training (minimizing reconstruction error)



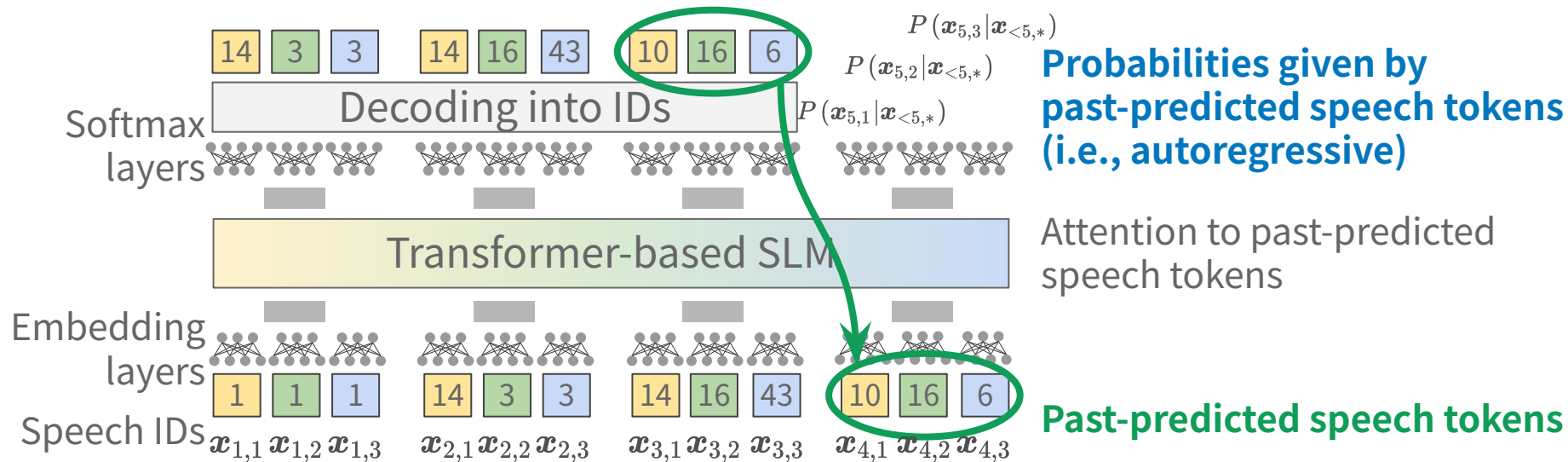
Speech RVQ encoder



Speech RVQ decoder



Modeling RVQ tokens (simplest implementation)

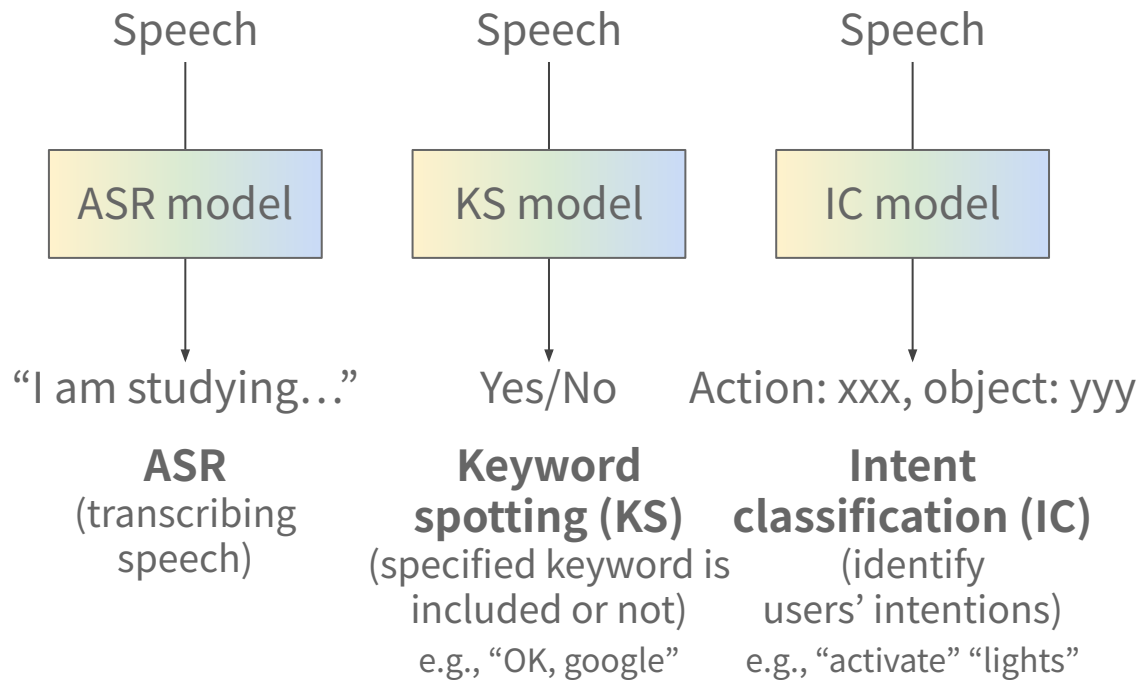


Case 2: speech modeling w/ task prompts

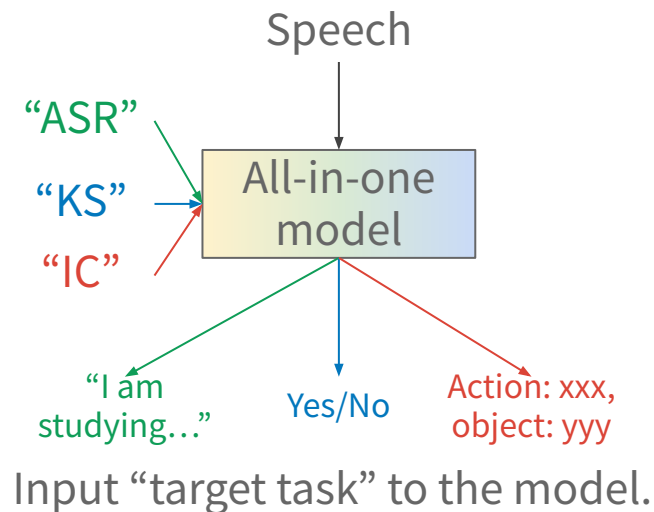
<https://arxiv.org/pdf/2203.16773> (SpeechPrompt)

Meaning of “task prompt”

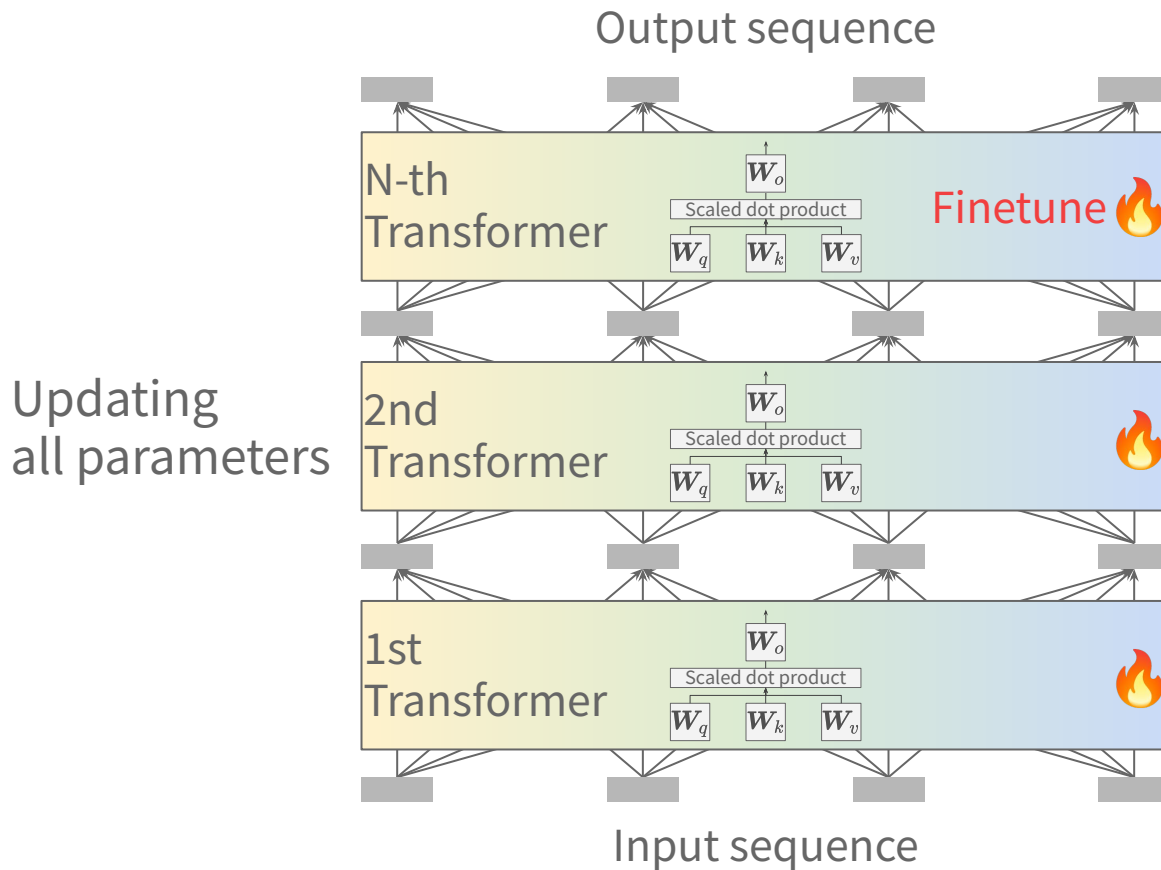
Full fine-tuning (task-specific model)



Task prompting (all-in-one model)

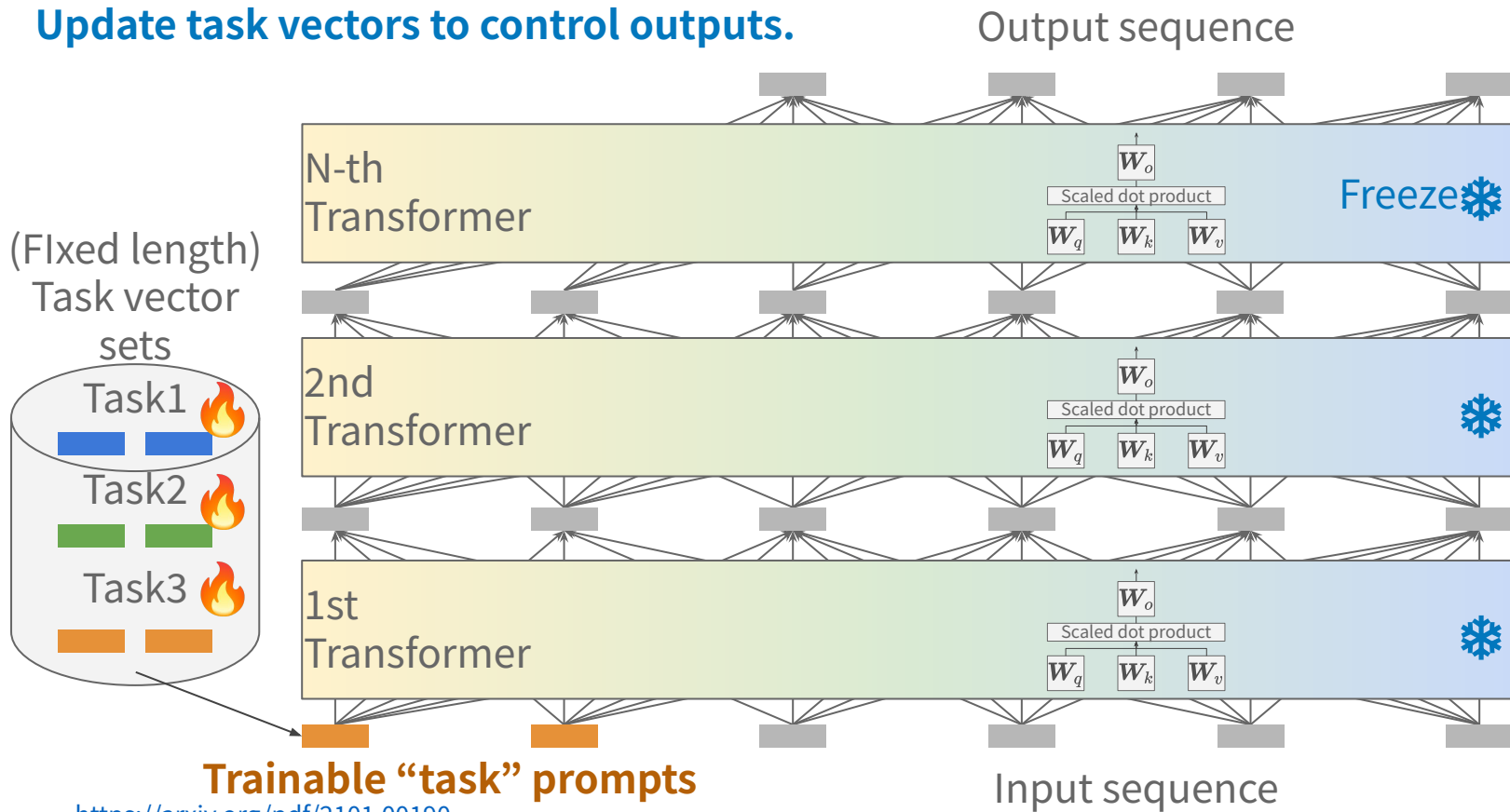


(Revisit) Finetuning Transformer decoder

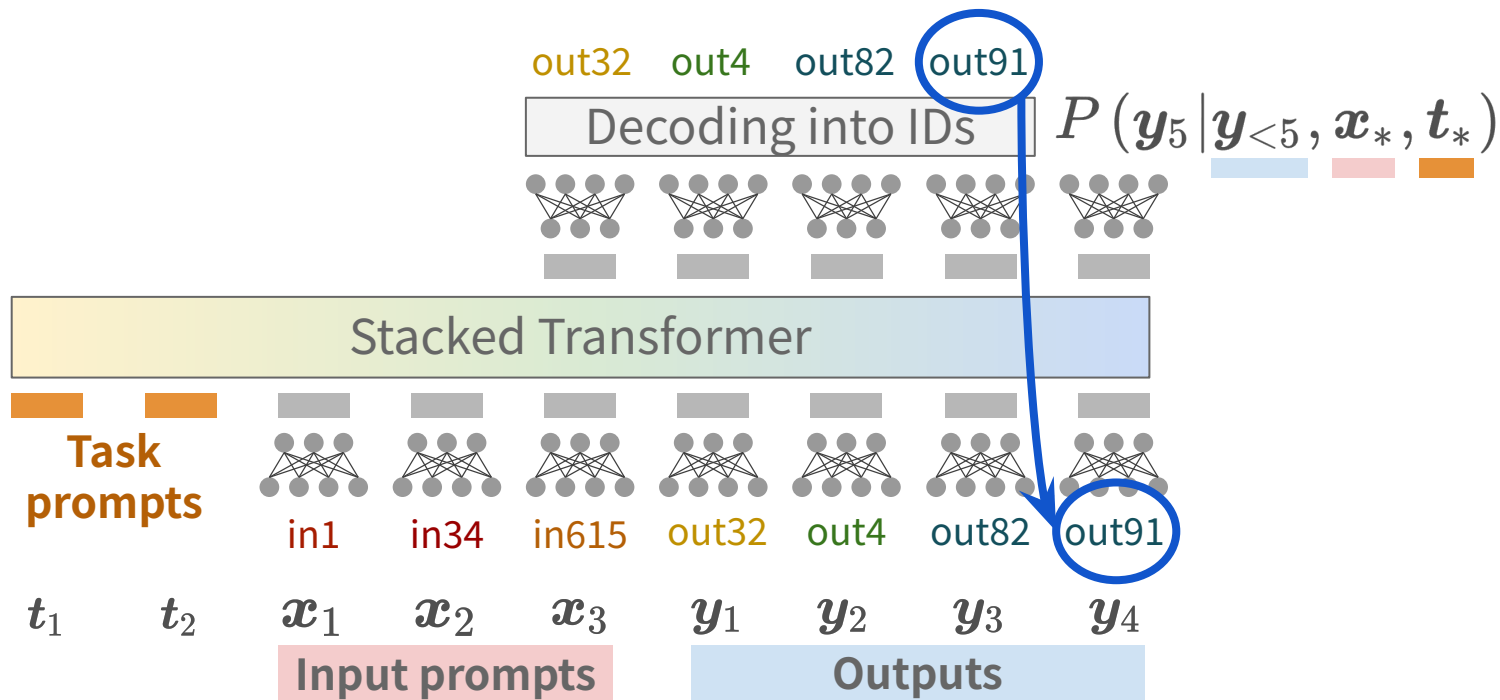


Prefix-Tuning: updating task-specific vectors

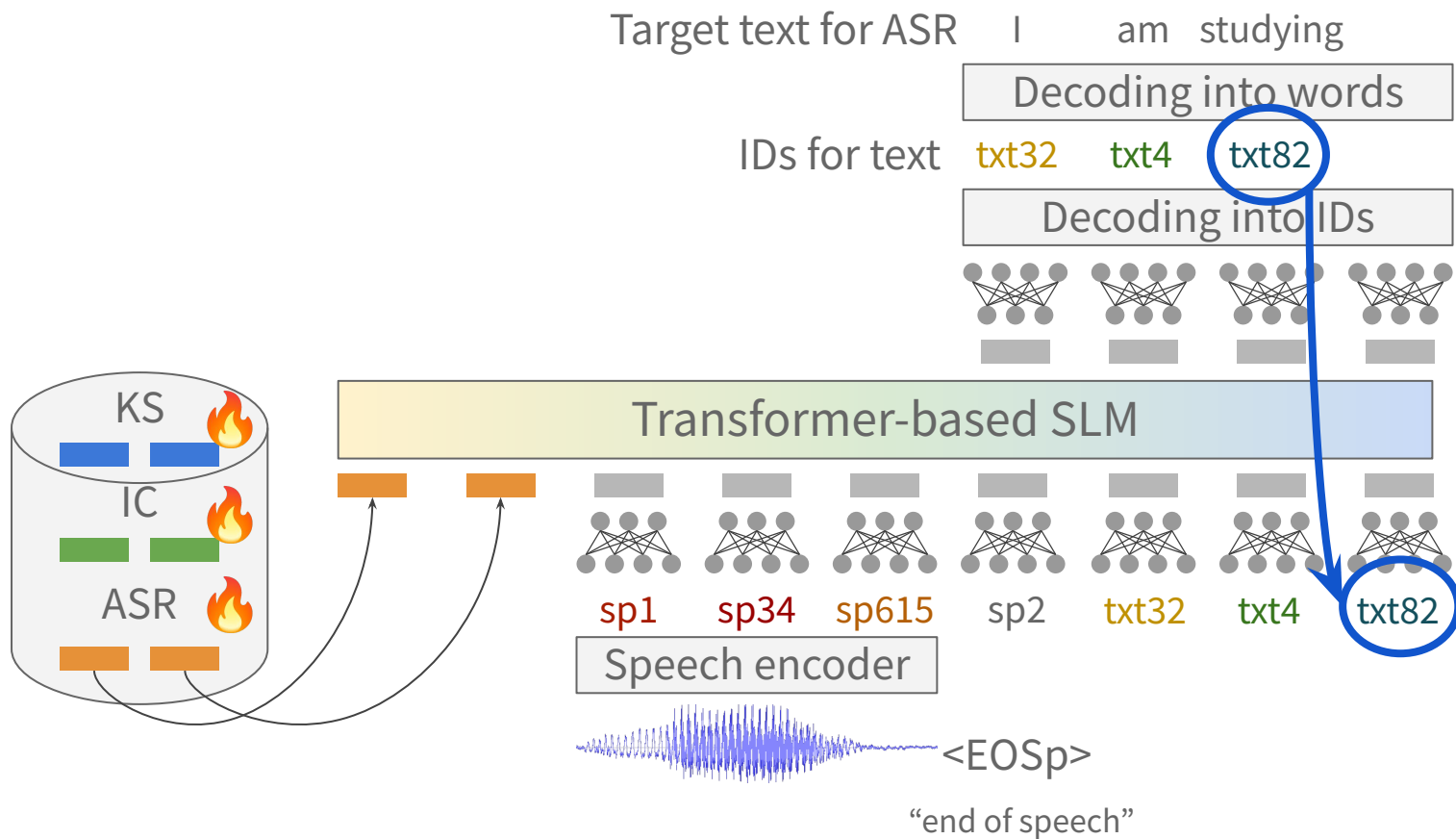
Update task vectors to control outputs.



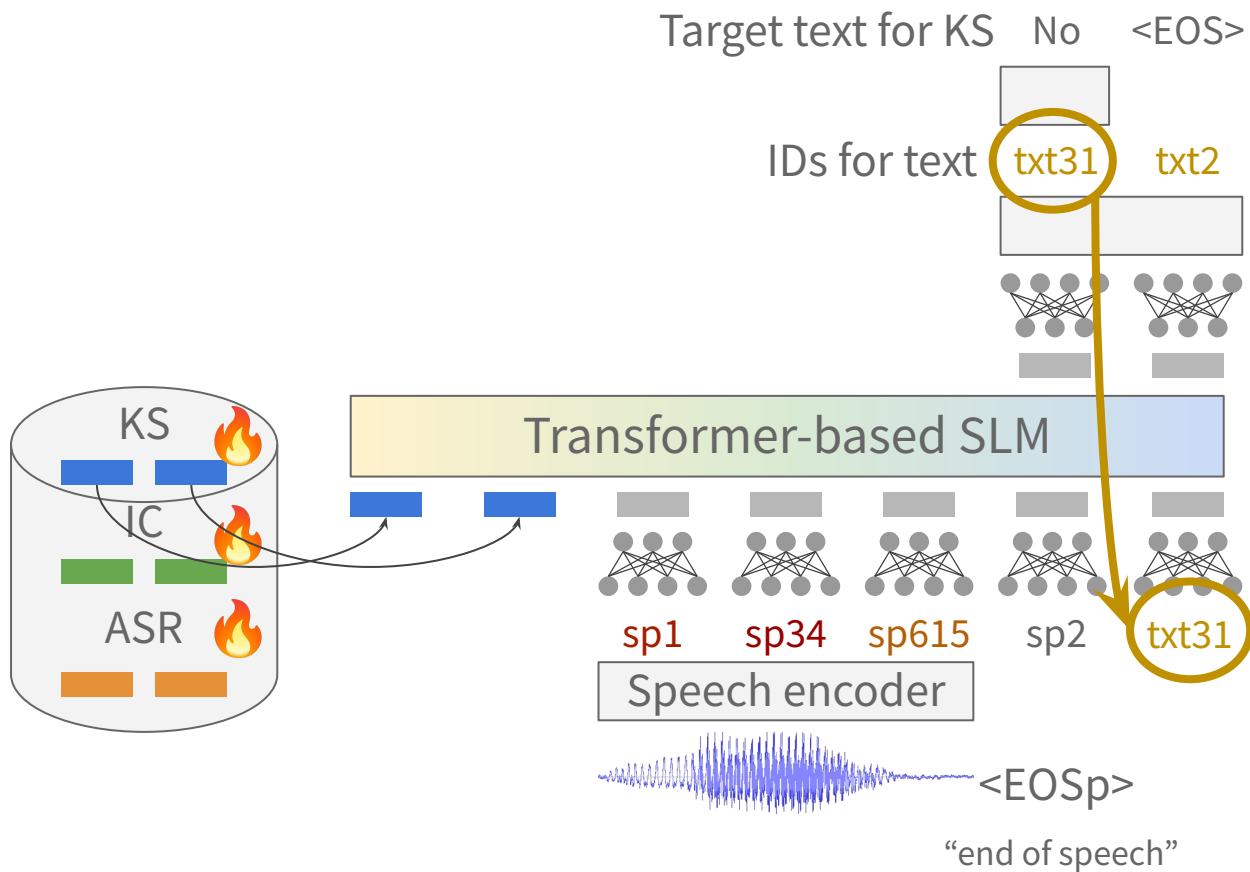
Inference: estimating output probability given by **task prompts**



Training SLM with task prompts (ASR)

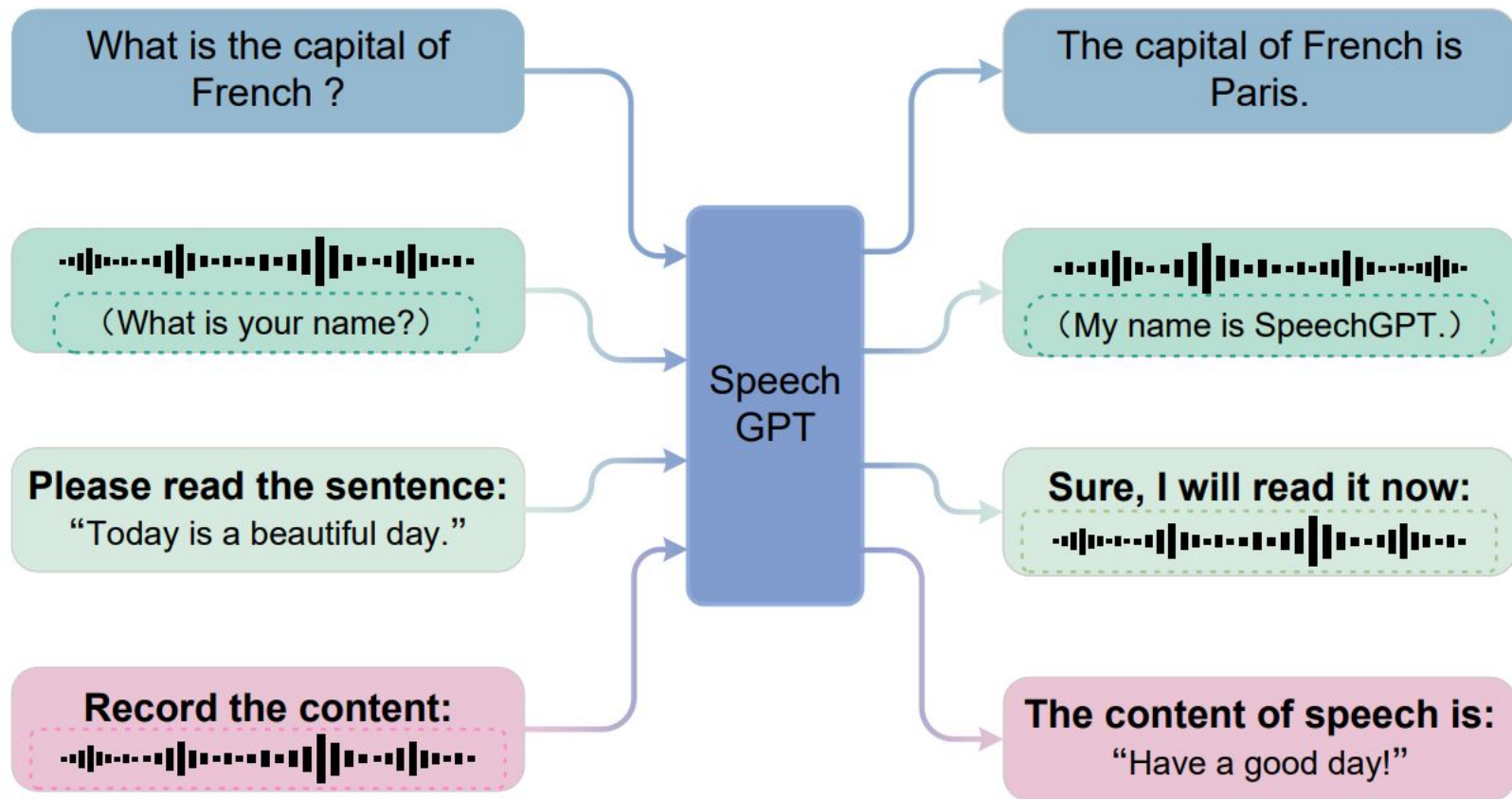


Training SLM with task prompts (KS, keyword spotting)

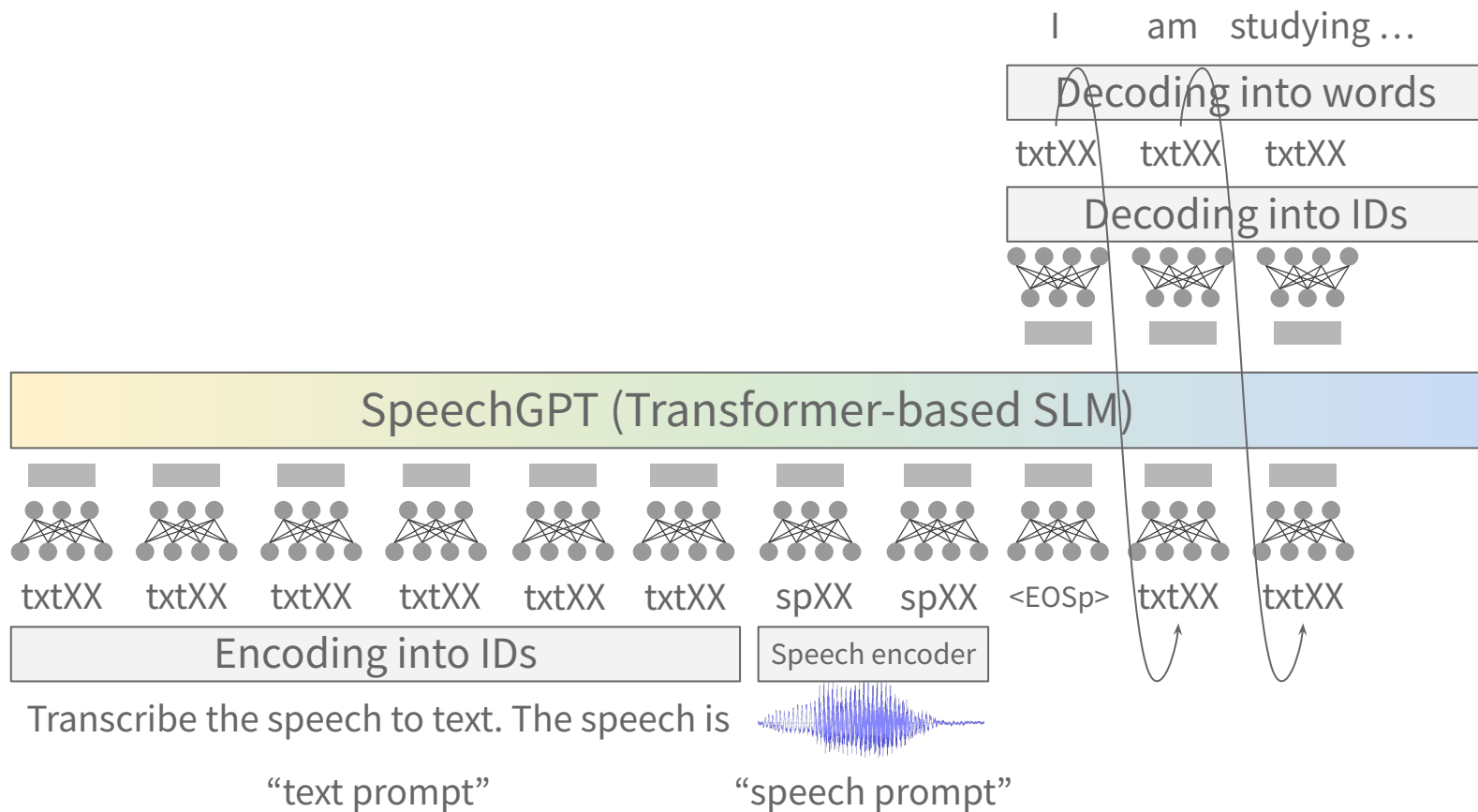


Case 3: speech modeling w/ text and speech prompts

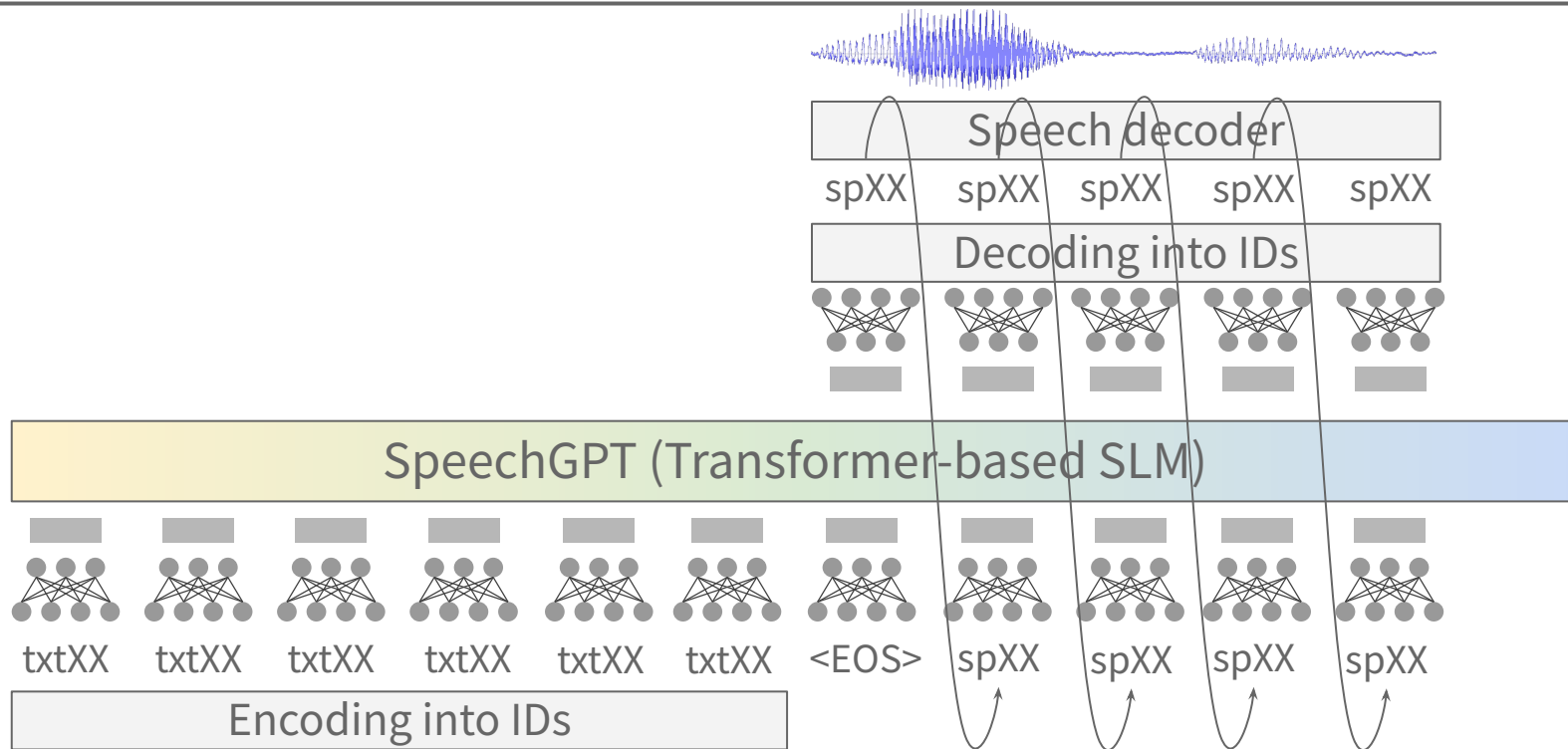
SpeechGPT: unifying speech-to-x and x-to-speech



SpeechGPT (speech-to-x)



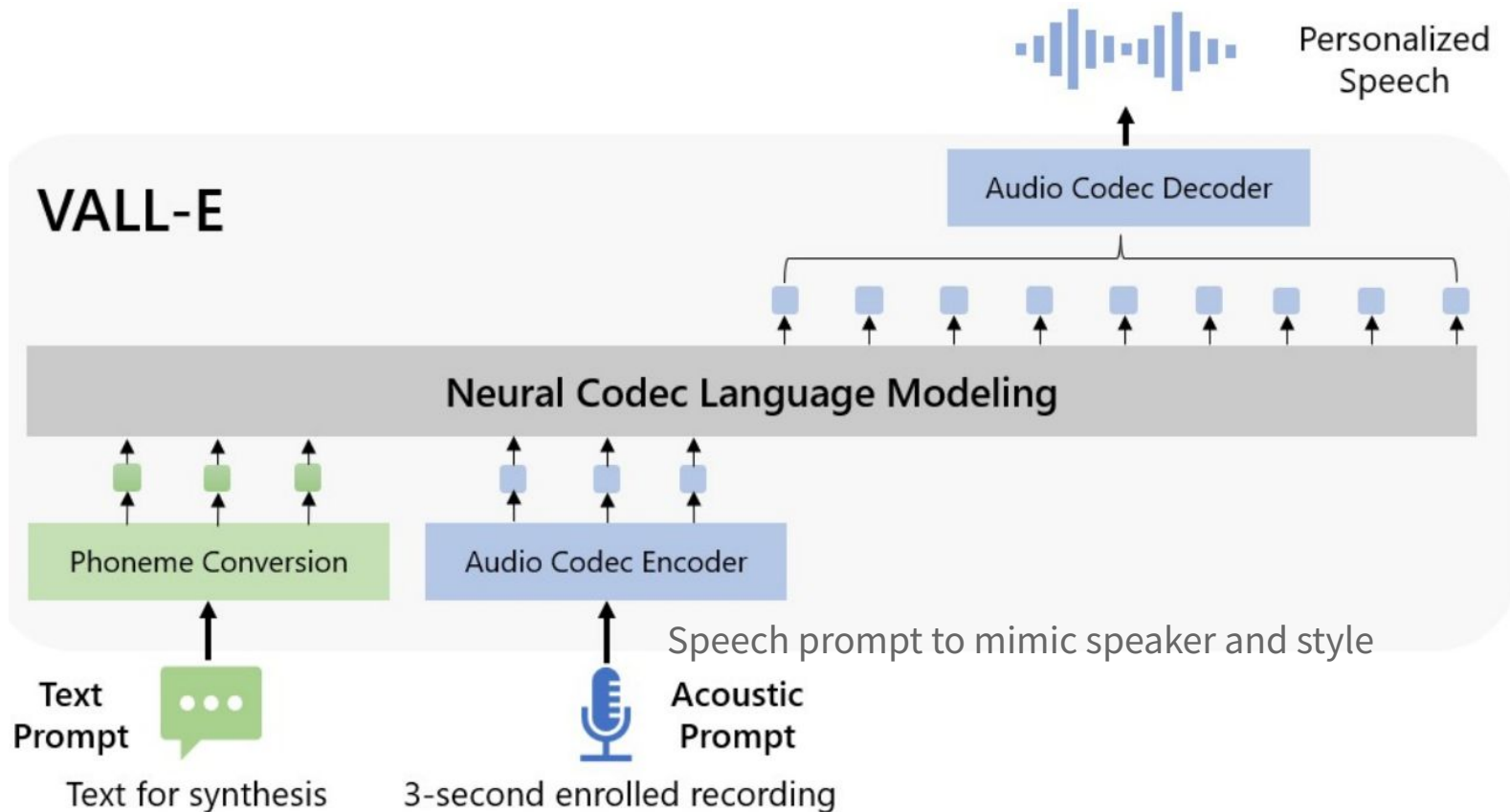
SpeechGPT (x-to-speech)



Please read the sentence: "Today is Sunday."

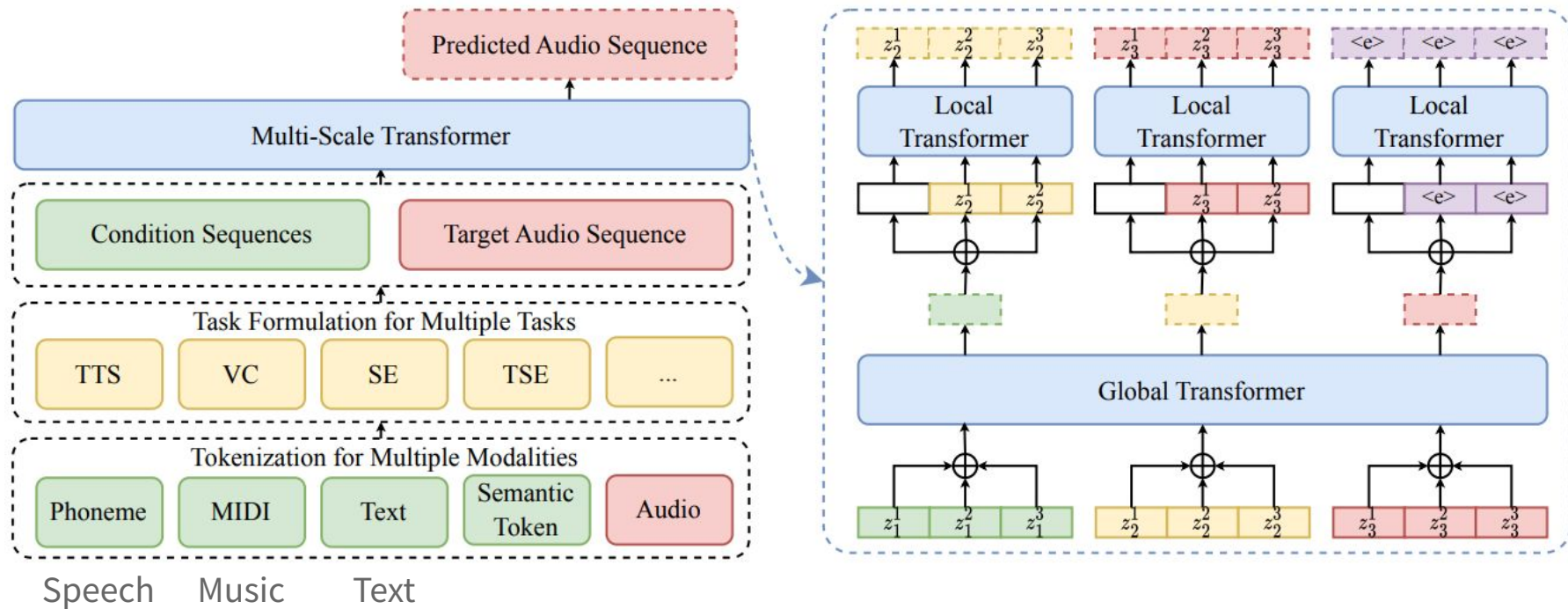
"text prompt"

VALL-E: SLM-based TTS



UniAudio: unifying speech, audio, and music tasks

Generating audio via “audio decoder”



Conclusion / まとめ

Conclusion

- Two meanings of foundation models
 - “Large pretrained model”
 - “All-in-one model”
- Large pretrained model
 - Finetuning, LoRA, MoE
- All-in-one model
 - Transformer-based speech modeling
 - Task prompts
 - Text and speech prompts

Next: special guest talk



Masahiro 水上 雅博 Mizukami, Ph.D.

Research Scientist.

Interaction Research Group

Innovative Communication Lab

NTT Communication Science Laboratories

Research Interests

対話システム

- Dialogue System (Chat-oriented, Task-oriented)
- Individuality Modeling
個人性モデリング